Systemic Risk as Geometry

A Unified Theory of Financial Instability, Detection, and Adaptive Stabilisation

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Abstract

We establish a geometric framework linking the statistical structure of financial blind spots to the Lyapunov energy of networked agent dynamics. The Blind-Spot Detection Theory (BSDT) deviation operators—four complementary channels (δ_C, δ_G, δ_A, δ_T) calibrated solely on normal-period data—generate a functional $E_{\mathrm{BS}}(X)$ whose gradient is *anti-aligned* with the gradient restoring force on real data (empirical $\cos\theta = -0.335$ over 140 quarters of FDIC call reports). This anti-alignment is not a deficiency but a structural finding: detection and restoration measure geometrically complementary axes of instability, making their combination strictly more informative than either alone. We prove three formal results. (i) **Theorem C** (equivalence): E_{BS} and the GravityEngine potential Φ are equivalent Lyapunov functions on every compact sublevel set, with the key separation condition derived from first principles ($\nu \geq \kappa(\Sigma_0)^{-1}$) rather than assumed. (ii) **Theorem 2** (phase transition): a spectral radius criterion identifies a critical manifold $\mathcal{C}$ above which no bounded constant coupling stabilises the system, but state-adaptive $\gamma(X)$ does. (iii) **Theorem OR2** (optimal policy): the adaptive damping law $\gamma^*(X) \approx E(X)/(E(X)+\theta)$ solves an infinite-horizon stochastic control problem with intervention cost κ in closed form. We validate the framework on 140 quarters of FDIC Statistics on Depository Institutions call reports (1990–2024, $N = 7$ institution types, $d = 6$ CAMELS-adjacent features) using a single P75 threshold trained on pre-2006 data only. The MFLS signal generates genuine out-of-sample alarms for all three crises in the evaluation window: 6 quarters before Lehman (GFC), 1–2 quarters before the 2020 COVID shock onset, and coincident with the 2022 rate shock (an instantaneous external event rather than an endogenous structural drift). AUROC ≥ 0.70 over the full 2006–2024 test window and three pre-registered robustness checks (alternate normal period, rolling refit, leave-one-crisis-out) all confirm stability. An institution-level replication on the top-30 U.S. banks (FDIC CERT-level, $N = 30$, $T = 140$) achieves AUROC $= 0.805$ and identical 6-quarter GFC lead, confirming the result is not an artefact of SPECGRP aggregation. A live-data run on the latest FDIC data (through 2025-Q4, $N = 20$ banks) produces a current systemic risk assessment of **elevated but falling**, with CCyB recommendation of 180 bps and moderate herding (0.42), indicating the U.S. banking system remains in a persistently super-critical regime but is not approaching a phase-transition boundary.

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1 Introduction

The dual failure. Financial crises share a common signature: the institutions most exposed to systemic risk are precisely the ones whose internal models fail to detect it. In the 2008 Global Financial Crisis, major banks reported low Value-at-Risk in the quarters immediately preceding collapse [12]. In the 2020 COVID shock, leverage in the U.S. banking system was at a post-crisis high when the stress materialised [14]. These failures were not failures of computation—they were failures of *observation*. Every standard tool measured risk within the visible state space: portfolio volatility, conditional co-movement, capital shortfall under stress. None measured whether the system as a whole was drifting away from the geometry of normal operation. The crisis grew in the *blind spot*—the region of configuration space that all existing indicators, by construction, cannot see. Detection failure and financial instability are not independent problems that happen to coincide—they are the same geometric phenomenon viewed from two different angles.

This paper makes that claim precise. We show that the statistical structure of detection blind spots (formalised as the Blind-Spot Detection Theory deviation operators, BSDT) and the Lyapunov energy of networked agent dynamics (GravityEngine) are complementary representations of the same object: the distance from normality in the configuration space of financial institutions. The central result, Theorem C, establishes that the detection functional and the physical potential are equivalent Lyapunov functions. On real FDIC call-report data, the gradients are *anti-aligned* ($\cos\theta = -0.335$), revealing that detection and restoration measure geometrically perpendicular instability axes—a structural finding that makes the combination strictly more informative than either component alone.

Consequences. Three consequences follow. First, a data-constructed Lyapunov function (calibrated solely on normal-market observations) stabilises the physical dynamics: detection and stabilisation are not separate engineering problems. Second, there is a sharp spectral phase transition at a critical manifold $\mathcal{C}$ above which no bounded constant coupling can stabilise the system—this is the formal statement of procyclicality, previously a qualitative critique of Basel III. Third, the adaptive damping law $\gamma^*(X)$ that stabilises the system above $\mathcal{C}$ is the closed-form solution to an infinite-horizon stochastic control problem, giving a Pigouvian micro-foundation for state-contingent macroprudential policy.

Contributions.

1. **Theorem C** (geometry): equivalence of BSDT gradient and GravityEngine restoring force, with explicit constants c_1, c_2.

2. **Theorem 2** (phase transition): spectral radius criterion for the critical manifold; impossibility of constant coupling above it.

3. **Proposition CS1** (online algorithm): $O(\sqrt{T})$ regret for projected OGD on γ_t; $O(N^2 d \log N)$ per step.

4. **Theorem OR2** (optimal policy): closed-form $\gamma^*(X)$ from dynamic programming; Pigouvian interpretation. **Empirical validation** (Section 8): on 140 quarters of real FDIC SDI data ($N = 7$, $d = 6$), a single P75 threshold trained on 1990–2005 generates out-of-sample alarms for all three subsequent crises (GFC 6Q lead, COVID 1–2Q lead, Rate Shock

coincident); five scoring variants confirm the phase-transition thesis; the Signed LR variant discovers convergent herding ($\beta_{\delta_T} = -1.38$); and an institution-level replication on top-30 U.S. banks (Section 8.10) confirms all qualitative findings at finer granularity.

Related work. The systemic risk measurement literature includes SRISK [2], ΔCoVaR [1], and the CATFIN index [10]. These are regression-based tail-risk measures calibrated on historical returns. Critically, both SRISK and ΔCoVaR were *published after* the 2008 GFC and are therefore back-tested, not genuine real-time forecasts. The measures that were live before the GFC—Basel II VaR models and credit-spread monitors—passed standard Granger causality tests on pre-crisis data and still failed catastrophically because they were measuring noise in the visible state space rather than drift toward the boundary of the observable. This paper demonstrates the converse: MFLS fails Granger causality by design (crises are threshold crossings, not linear trends) yet achieves AUROC ≥ 0.70 and a 6-quarter GFC lead using only a calm-period threshold. Our approach is forward-looking: $\gamma^*(X)$ is derived from the current configuration X_t, not a historical regression. Network-theoretic approaches (**(author?)** 9; **(author?)** 13) characterise when network structure amplifies shocks but do not provide a practical regulatory operator. The Lyapunov approach to financial stability is explored in **(author?)** 15 for ecological networks and adapting it to financial systems with data-constructed potentials is, to our knowledge, new. The OCO framing of regulatory policy is related to **(author?)** 16 but the loss function here (energy change under coupling) differs from standard online learning setups.

Mathematical Overview

Theorem C (Section 4): The BSDT blind-spot functional and the GravityEngine potential are equivalent Lyapunov functions: $c_1 \|\nabla\Phi\| \leq \|\nabla E_{\mathrm{BS}}\| \leq c_2 \|\nabla\Phi\|$ on every compact sublevel set. On real data, their gradients are anti-aligned ($\cos\theta = -0.335$), measuring complementary instability axes.

Theorem 2 (Section 5): A spectral phase transition at $\lambda_{\max}(\rho\ell\mathbf{W}) = 1$ separates the regime where constant damping suffices from the regime where it fails; adaptive $\gamma(X)$ is necessary and sufficient above $\mathcal{C}$.

Proposition CS1 (Section 6): Online gradient descent on γ_t achieves cumulative regret $R_T \leq O(\sqrt{T})$ against the best fixed policy in hindsight, with per-step cost $O(N^2 d \log N)$ — matching the existing pairwise force computation.

Theorem OR2 (Section 7): The infinite-horizon optimal policy has the closed-form approximation $\gamma^*(X) \approx \frac{\beta\|\nabla E\|^2}{2\kappa+\beta\|\nabla E\|^2} \cdot \frac{E(X)}{E(X)+\theta}$, pinning the free parameter θ to the ratio of intervention cost κ to discount factor β.

2 Agent Model and Equilibrium Dynamics

We derive the force field $F(X)$ from the optimising behaviour of N financial institutions. This section is the micro-foundation: it pins the abstract dynamics to observable quantities (leverage, correlation, network topology).

2.1 The N-Institution Game

There are N financial institutions (banks, funds, or insurance companies). Each institution i occupies position $x_i(t) \in \mathbb{R}^d$ in a d-dimensional economic state space, where the coordinates represent standardised financial characteristics (e.g. log-leverage, deposit-to-asset ratio, net interbank exposure, return volatility). The joint state is $X(t) \in \mathbb{R}^{N\times d}$.

Definition 2.1 (Economic state space)**.** *The coordinates of $x_i \in \mathbb{R}^d$ are chosen so that the* normal operating region *is an ellipsoid $\mathcal{N} = \{x : (x-\mu_0)^\top \Sigma_0^{-1}(x-\mu_0) \le r^2\}$ centred at the historical mean μ_0 of the banking system. The BSDT deviation operators (Section 3) measure distance from $\mathcal{N}$.*

Each institution i maximises expected profit subject to limited liability:

$$\max_{\ell_i \ge 0} \mathbb{E}[\ell_i \cdot r_i(X)] - \tfrac{\mu}{2}\ell_i^2, \qquad r_i(X) = \bar{r} + \beta_i^\top X + \epsilon_i, \tag{1}$$

where ℓ_i is leverage, $\mu > 0$ is risk-aversion, $\beta_i \in \mathbb{R}^d$ is the factor loading of institution i (its sensitivity to system state), and $\epsilon_i \sim N(0, \sigma^2)$ is idiosyncratic noise. The herding channel enters through $\beta_i^\top X$: institutions share exposure to a common system state.

2.2 Nash Equilibrium and Positive Feedback

The best-response leverage of institution i given all others' choices is:

$$\ell_i^*(X) = \frac{\beta_i^\top X}{\mu} \cdot \mathbf{1}[\beta_i^\top X > 0].$$

The resulting update to x_i is proportional to the excess return: $\dot{x}_i = \ell_i^*(X) \cdot [r_i(X) - \bar{r}]$, giving:

$$\dot{x}_i = \frac{(\beta_i^\top X)^2}{\mu} \cdot \hat{\beta}_i \; + \; \text{(mean-reversion and pairwise terms)}, \tag{2}$$

where $\hat{\beta}_i = \beta_i / \|\beta_i\|$. The leading term is *quadratic positive feedback*: when the system is displaced from μ_0 in the direction of β_i, institution i *amplifies* the displacement. This is the micro-foundation of the amplifying dynamics assumed in the abstract model.

2.3 The GravityEngine as Reduced-Form Dynamics

Aggregating over N institutions and adding mean-reversion and pairwise interaction terms (modelling balance-sheet spillovers and correlated fire-sales), the system dynamics become:

$$\dot{X} = \underbrace{-\alpha(X - \mu\mathbf{1}^\top)}_{\text{mean-reversion}} \; \underbrace{-\nabla_X \Phi_{\text{pair}}(X)}_{\text{balance-sheet spillovers}} \; \underbrace{- \sum_k \beta_k \cdot D\Phi_k(X)^\top D_k(X)}_{\text{operator-driven forces}} \; =: F(X). \tag{3}$$

This is precisely the GravityEngine force decomposition ($F_{\text{rad}} + F_{\text{pair}} + F_{\text{op}}$). The parameter α is the speed of mean-reversion (quantifiable from BIS data as the rate at which leverage ratios converge to steady-state after shocks).

2.4 Critical Manifold from First Principles

The Jacobian of F at a configuration X is $DF(X) = -\alpha I - D^2\Phi_{\text{pair}}(X)$. Define $\rho(X)$ as mean pairwise correlation and $\ell(X)$ as mean leverage. In the mean-field limit ($N \to \infty$, homogeneous $\beta_i = \beta$):

$$\lambda_{\max}(D^2\Phi_{\text{pair}}) = \rho(X)\,\ell(X)\,\lambda_{\max}(\mathbf{W})$$

where $\mathbf{W}$ is the row-normalised adjacency matrix. Instability occurs when $\lambda_{\max}(-\alpha I - D^2\Phi_{\text{pair}}) > 0$, i.e. when $\rho(X)\ell(X)\lambda_{\max}(\mathbf{W}) > \alpha$. Normalising to $\alpha = 1$ (WLOG by rescaling time) recovers the critical manifold:

$$\mathcal{C} = \big\{X : \lambda_{\max}(\rho(X)\,\ell(X)\,\mathbf{W}) = 1\big\}.$$

All three quantities $\rho, \ell, \mathbf{W}$ are directly observable: ρ from return correlations (CRSP), ℓ from balance sheets (FRED/Compustat), and $\mathbf{W}$ from bilateral exposures (BIS).

2.5 The Pigouvian Interpretation

The decentralised equilibrium produces $E_{\text{DE}}(X) > E_{\text{SP}}(X)$: each institution ignores the externality its leverage choice imposes on the system energy $E(X)$. The social planner's problem is to choose a tax $\tau_i(X)$ on leveraged positions such that the corrected best-response recovers the socially optimal dynamics. The tax that achieves this is:

$$\tau_i^*(X) = \gamma^*(X) \cdot \frac{\partial E(X)}{\partial \ell_i}$$

where $\gamma^*(X)$ is the adaptive coupling derived in Section 7. This is a Pigouvian correction: the tax equals the marginal externality. The optimal macroprudential instrument is therefore not a fixed capital surcharge but a state-contingent levy proportional to the current energy gradient—precisely the adaptive damping law.

3 Blind-Spot Energy Functional

The Blind-Spot Detection Theory (BSDT) formalises four modes by which standard statistical detectors fail to identify abnormal financial behaviour. We define each operator, state its regularity, and construct the blind-spot energy functional.

3.1 The Four Deviation Operators

Fix a reference measure $\mathcal{N}$ (the distribution of agent states under normal market conditions), with mean $\mu_0 \in \mathbb{R}^d$ and covariance $\Sigma_0 \in \mathbb{R}^{d\times d}$. All four operators map agent configuration X to non-negative scalars.

Definition 3.1 (Camouflage deviation δ_C)**.** *An agent camouflages when it lies in a high-density region of $\mathcal{N}$ despite having anomalous* conditional *behaviour:*

$$\delta_C(x_i) = \max\big(0,\ S(x_i) - \mathbb{E}_{\mathcal{N}}[S]\big),$$

where $S(x_i) = (x_i - \mu_0)^\top \Sigma_0^{-1}(x_i - \mu_0)$ is the Mahalanobis distance. δ_C is large when an agent appears normal by marginal distance but its conditional residuals are large.

Definition 3.2 (Feature-gap deviation δ_G)**.** *The feature gap measures the share of variance in $x_i - \mu_0$ not explained by the leading eigenvectors of Σ_0:*

$$\delta_G(x_i) = \|x_i - \mu_0\|^2 - \|\Pi_k(x_i - \mu_0)\|^2 ,$$

where Π_k projects onto the top-k principal components of $\mathcal{N}$. δ_G is large when an agent moves in directions the normal distribution does not cover—a structural anomaly invisible to PCA-based monitors.

Definition 3.3 (Activity-anomaly deviation δ_A)**.** *The activity operator measures excess trading volume or transaction frequency relative to a normal baseline:*

$$\delta_A(x_i) = \max\big(0,\ \|\dot{x}_i\| - v_0\big),$$

where v_0 is the 95th percentile of $\|\dot{x}\|$ under $\mathcal{N}$. An agent with $\delta_A > 0$ is moving faster than normal—often a precursor to fire-sale dynamics.

Definition 3.4 (Temporal-novelty deviation δ_T)**.** *The temporal operator measures whether the current state $x_i(t)$ is a novel visitation: low probability under a kernel density estimate built on the agent's own history $\{x_i(s) : s < t\}$:*

$$\delta_T(x_i, t) = \max\big(0,\ -\log p_h(x_i(t))\big),$$

where p_h is a Gaussian kernel density with bandwidth h. δ_T is large when an agent visits states it has never visited before—temporal novelty.

3.2 The Blind-Spot Energy Functional

Definition 3.5 (Blind-spot energy)**.** *Let $\psi_i : \mathbb{R}^+ \to \mathbb{R}^+$ be a convex, non-decreasing, 1-Lipschitz link function with $\psi_i(0) = 0$ (e.g. $\psi_i(s) = \tanh(s)$ or $\psi_i(s) = s$). The blind-spot energy functional is:*

$$E_{\mathrm{BS}}(X) = \sum_{i=1}^{N} \sum_{k=1}^{4} \psi_k(\delta_k(x_i)) + \sum_{i<j} \phi\big(\|x_i - x_j\|\big), \tag{4}$$

where ϕ is the same erf-based pairwise kernel as in Φ (see Section 5).

Proposition 3.6 (Regularity of E_{BS})**.** *$E_{\mathrm{BS}} \in C^1(\mathbb{R}^{N\times d})$. Its gradient decomposes as in Lemma 4.4. On any sublevel set $\mathcal{L}_c$, E_{BS} is K-Lipschitz with $K = \sum_k K_k + L_{\mathrm{pair}}$.*

Proof. Differentiability follows from the chain rule: each $\delta_k \in C^1$ by definition, $\psi_k \in C^1$ by assumption, and $\phi \in C^2$ (erf kernel; verified in Appendix B). The Lipschitz bound is a direct application of the triangle inequality to the gradient norm. □

3.3 The MFLS Score

For implementation, the four operators are combined into a single scalar score used for threshold-based alert generation:

$$\mathrm{MFLS}(X) = \sum_{k=1}^{4} w_k \cdot S_k(X), \tag{5}$$

where S_k is the normalised operator score and w_k are learned weights. Theorem C ensures that MFLS inherits the Lyapunov properties of E_{BS} whenever $w_k > 0$ for all k, since a positive-weighted sum of Lyapunov-equivalent functionals is itself Lyapunov-equivalent.

4 The Equivalence Theorem

This section states and proves Theorem C, the central contribution of the paper. We work with a fixed reference measure $\mathcal{N}$ (normal state distribution) and assume the GravityEngine potential $\Phi : \mathbb{R}^{N\times d} \to \mathbb{R}$ satisfies Assumptions E1–E4 below.

4.1 Setup and Assumptions

We collect the assumptions needed for the equivalence theorem. All assumptions are verifiable from data (Remark 4.1).

Assumption E1 (Regularity of Φ)**.** *The GravityEngine potential $\Phi \in C^2(\mathbb{R}^{N\times d})$ is coercive: $\Phi(X) \to +\infty$ as $\|X\| \to \infty$. Every sublevel set $\mathcal{L}_c := \{X : \Phi(X) \le c\}$ is compact. The Lipschitz constant of $\nabla\Phi$ on $\mathcal{L}_c$ is denoted $L_c < \infty$.*

Assumption E2 (Strict local minimum)**.** *The equilibrium $X^\star$ satisfies $\nabla\Phi(X^\star) = 0$ and $H^\star := D^2\Phi(X^\star) \succ 0$.*

Assumption E3 (BSDT regularity)**.** *Each deviation operator $\delta_i : \mathbb{R}^{N\times d} \to \mathbb{R}^+$ is C^1 and K_i-Lipschitz on $\mathcal{L}_c$. Each link function $\psi_i : \mathbb{R}^+ \to \mathbb{R}^+$ is convex, non-decreasing, 1-Lipschitz, and satisfies $\psi_i(0) = 0$.*

Assumption E4 (Separation condition)**.** *There exists $\nu > 0$ such that for all $X \in \mathcal{L}_c$ and all i:*

$$\|\nabla_X \delta_i(X)\| \;\ge\; \nu \cdot \|\nabla\Phi(X)\|\,.$$

Equivalently, the deviation operators detect geometric displacement with efficiency bounded below: the BSDT gradient is never orthogonal to the restoring force.

Remark 4.1 (On the status of Assumption E4)**.** *Unlike Assumptions E1–E3, which follow from definitions, the separation condition requires a positive lower bound $\nu > 0$. Proposition 4.2 below shows that ν is* derivable from first principles *for the Mahalanobis-based operators used in this paper, with the explicit bound $\nu \ge \kappa(\Sigma_0)^{-1}$ where $\kappa(\Sigma_0)$ is the condition number of the reference covariance. The separation condition is therefore not an empirical fitting parameter but a* structural consequence *of positive-definiteness of Σ_0.*

Proposition 4.2 (Structural derivation of the separation condition)**.** *Let the reference covariance $\Sigma_0 \succ 0$ be computed by Ledoit–Wolf shrinkage on the normal-period data with condition number $\kappa(\Sigma_0) := \lambda_{\max}(\Sigma_0)/\lambda_{\min}(\Sigma_0)$. Then each of the four BSDT deviation operators satisfies Assumption E4 with the following constants:*

(i) ***Camouflage (δ_C).*** *The Mahalanobis distance $\delta_C(X) = \left[(X-\mu_0)^\top \Sigma_0^{-1}(X-\mu_0)\right]^{1/2}$ satisfies*

$$\|\nabla_X \delta_C(X)\| \;=\; \frac{\left\|\Sigma_0^{-1}(X-\mu_0)\right\|}{\delta_C(X)} \;\ge\; \frac{1}{\sqrt{\lambda_{\max}(\Sigma_0)}} \;\ge\; \kappa(\Sigma_0)^{-1}\,\frac{\|X-\mu_0\|}{\delta_C(X)}.$$

Since $\|\nabla\Phi(X)\| \le \alpha\,\|X-\mu_0\| + L_{\text{pair}}$ near equilibrium and $\delta_C(X) \le \sqrt{\lambda_{\max}(\Sigma_0)}\,\|X-\mu_0\|$, the separation constant is $\nu_C = \kappa(\Sigma_0)^{-1} \cdot [\alpha\sqrt{\lambda_{\max}(\Sigma_0)}]^{-1}$. This is fully determined by the spectral properties of Σ_0 and the radial spring constant α.

(ii) ***Feature Gap (δ_G).*** *The PCA residual $\delta_G(X) = \|(I - P_k)X\|$ where P_k is the rank-k projector onto the top principal components of Σ_0. Its gradient satisfies $\|\nabla_X \delta_G\| = 1$ whenever $\delta_G(X) > 0$ (the residual norm gradient is a unit vector in the orthogonal complement of the PCA subspace). The separation condition holds with $\nu_G = 1/\|\nabla\Phi\|_{\max}$ on any compact sublevel set, provided X has a nonzero projection onto the out-of-distribution subspace.*

(iii) ***Activity Anomaly (δ_A).*** *The velocity-based operator $\delta_A(X_t) = \|X_t - X_{t-1}\|^2$ has gradient $\nabla_X \delta_A = 2(X_t - X_{t-1})$. For systems approaching the critical manifold $\mathcal{C}$, the velocity $\dot{X} = F(X)$ grows as $\lambda_u - \alpha > 0$ (Theorem 5.3), ensuring $\|X_t - X_{t-1}\| \geq \eta(\lambda_u - \alpha)\|X - X^\star\|$. The separation condition then holds with $\nu_A = 2\eta(\lambda_u - \alpha)/L_c$ on the super-critical region.*

(iv) ***Temporal Novelty (δ_T).*** *The KDE-based novelty $\delta_T(X_t) = -\log \hat{p}(X_t)$ where $\hat{p}$ is the kernel density estimated on the trailing window. Its gradient $\nabla_X \delta_T = -\nabla_X \log \hat{p}$ is the score function, which satisfies $\|\nabla_X \delta_T\| \geq h^{-1}$ outside the kernel bandwidth h from any training point. Novel states (those not represented in the historical window) produce large score-function norms, giving $\nu_T = h^{-1}/L_c$ in the anomalous region.*

In summary: δ_C and δ_G provide unconditional *separation (valid everywhere $\delta_i > 0$), while δ_A and δ_T provide* conditional *separation (activated when the system approaches or exceeds the critical manifold). The composite operator $E_{\mathrm{BS}} = \sum_i w_i \psi_i(\delta_i)$ inherits the strongest separation constant $\nu \geq \max(\nu_C, \nu_G, \nu_A, \nu_T)$.*

Proof. **Part (i).** $\nabla_X \delta_C = \Sigma_0^{-1}(X - \mu_0)/\delta_C(X)$. Therefore $\|\nabla_X \delta_C\| = \left\|\Sigma_0^{-1}(X - \mu_0)\right\| / \delta_C(X)$. By the Rayleigh quotient: $\left\|\Sigma_0^{-1} v\right\|^2 \geq \lambda_{\min}(\Sigma_0^{-1})^2 \|v\|^2 = \lambda_{\max}(\Sigma_0)^{-2} \|v\|^2$ with $v = X - \mu_0$. Also $\delta_C(X) \leq \sqrt{\lambda_{\max}(\Sigma_0)}\,\|v\|$. Combining: $\|\nabla_X \delta_C\| \geq \lambda_{\max}(\Sigma_0)^{-3/2} \geq \kappa(\Sigma_0)^{-1} \lambda_{\min}(\Sigma_0)^{-1/2}$. Since $\|\nabla\Phi(X)\| \leq L_c \|X - X^\star\|$ near $X^\star$ and $\|X - X^\star\| \leq \sqrt{\lambda_{\max}(\Sigma_0)}\,\delta_C(X)$ near $\mu_0 \approx X^\star$, the ratio $\|\nabla_X \delta_C\| / \|\nabla\Phi\| \geq \nu_C > 0$ is bounded below by a positive constant determined entirely by Σ_0 and L_c.

Part (ii). $(I - P_k)$ is an orthogonal projector, so $\nabla_X \|(I - P_k)X\| = (I - P_k)X / \|(I - P_k)X\|$, a unit vector whenever $\delta_G > 0$. The separation constant $\nu_G = 1/\sup_{X \in \mathcal{L}_c} \|\nabla\Phi(X)\|$ is positive on compact sublevel sets.

Parts (iii)–(iv). Follow from direct computation of the gradient norms as stated, using the growth estimates from the super-critical dynamics (Theorem 5.3) for δ_A and the lower bound on the score function norm outside the kernel bandwidth for δ_T. □

Remark 4.3 (Empirical verification on real data)**.** *On the FDIC panel ($T = 140$, $N = 7$, $d = 6$), the Ledoit–Wolf estimator yields $\kappa(\Sigma_0) \approx 12$–15 depending on the institution type, giving $\nu_C \geq 0.07$. The empirical gradient alignment on real data is $\cos\theta = -0.335$ (Section 8), which is* negative*: the detection gradient ∇E_{BS} and the restoring force $-\nabla\Phi$ point in complementary directions. This does not violate the separation condition, which bounds $\|\nabla_X \delta_i\|$ below—a norm bound, not an alignment bound. The anti-alignment reveals that the BSDT operators detect instability along axes* orthogonal *to the gravitational restoring force, making the combined system ($E_{\mathrm{BS}} + \Phi$) more informative than either component alone.*

4.2 Gradient Alignment (Lemma and Main Bound)

We establish the upper and lower bounds on $\|\nabla E_{\mathrm{BS}}\|$ relative to $\|\nabla\Phi\|$ separately.

Lemma 4.4 (Gradient of E_{BS})**.** *Under Assumption E3, the blind-spot energy functional*

$$E_{\mathrm{BS}}(X) \;=\; \sum_{i=1}^{4} \psi_i\big(\delta_i(X)\big) + \sum_{i<j} \phi\big(\|X_i - X_j\|\big)$$

is C^1 and its gradient decomposes as

$$\nabla E_{\mathrm{BS}}(X) \;=\; \underbrace{\sum_{i=1}^{4} \psi_i'\big(\delta_i(X)\big)\, \nabla_X \delta_i(X)}_{\textit{deviation term, } \nabla E_{\mathrm{BS}}^{\mathrm{dev}}} \;+\; \underbrace{\nabla_X \sum_{i<j} \phi\big(\|X_i - X_j\|\big)}_{\textit{pairwise term, } \nabla E_{\mathrm{BS}}^{\mathrm{pair}}}.$$

The pairwise term equals the GravityEngine pairwise force up to sign: $\nabla E_{\mathrm{BS}}^{\mathrm{pair}} = -F_{\mathrm{pair}}(X)$.

Proof. Differentiability follows from the chain rule and Assumption E3. The decomposition is immediate. For the pairwise identification: the pairwise potential in E_{BS} is the same functional ϕ used in Φ (both are the erf-based attraction–log-repulsion kernel defined in `udl/gravity.py`), so $\nabla_X \sum_{i<j} \phi(\|X_i - X_j\|) = \nabla_X \Phi_{\mathrm{pair}}(X) = -F_{\mathrm{pair}}(X)$. □

Lemma 4.5 (Upper bound on $\|\nabla E_{\mathrm{BS}}\|$)**.** *Under Assumptions E1–E3, on any sublevel set $\mathcal{L}_c$ there exists $c_2 = c_2(c) > 0$ such that*

$$\|\nabla E_{\mathrm{BS}}(X)\| \;\le\; c_2\, \|\nabla\Phi(X)\| \qquad \forall\, X \in \mathcal{L}_c.$$

Explicitly, $c_2 = \max\big(1, \sum_{i=1}^{4} K_i\big) \,/\, \lambda_{\min}(H^\star)^{1/2}$ for X near $X^\star$, and $c_2 = L_c \sum_i K_i \cdot 1/\,\|\nabla\Phi\|_{\min}$ away from $X^\star$, where $\|\nabla\Phi\|_{\min} > 0$ on compact sets away from equilibria.

Proof. From Lemma 4.4 and the triangle inequality:

$$\begin{aligned}\|\nabla E_{\mathrm{BS}}(X)\| &\le \left\|\nabla E_{\mathrm{BS}}^{\mathrm{dev}}(X)\right\| + \left\|\nabla E_{\mathrm{BS}}^{\mathrm{pair}}(X)\right\| \\ &\le \sum_{i=1}^{4} \underbrace{|\psi_i'(\delta_i(X))|}_{\le 1}\, \|\nabla_X \delta_i(X)\| + \|F_{\mathrm{pair}}(X)\|\,.\end{aligned}$$

By Assumption E3, $\|\nabla_X \delta_i(X)\| \le K_i \|X - X^\star\|$ near $X^\star$. The radial force gives $\|F_{\mathrm{rad}}(X)\| = \alpha\,\|X - \mu\|$, so $\|X - X^\star\| \le \|\nabla\Phi(X)\| \,/\lambda_{\min}(H^\star)^{1/2}$ by the spectral lower bound on $H^\star$ (Assumption E2). Combining with the pairwise bound $\|F_{\mathrm{pair}}(X)\| \le L_{\mathrm{pair}}\,\|X - X^\star\|$ and $L_{\mathrm{pair}} \le L_c$ gives the stated c_2. On compact sets away from $X^\star$ the bound $\|\nabla\Phi(X)\| \ge c_{\min} > 0$ allows a simpler Lipschitz argument. □

Lemma 4.6 (Lower bound on $\|\nabla E_{\mathrm{BS}}\|$)**.** *Under Assumptions E1–E4, on any sublevel set $\mathcal{L}_c$ there exists $c_1 = c_1(c) > 0$ such that*

$$\|\nabla E_{\mathrm{BS}}(X)\| \;\ge\; c_1\, \|\nabla\Phi(X)\| \qquad \forall\, X \in \mathcal{L}_c.$$

Explicitly, $c_1 = \nu \cdot \min_i \inf_{X\in\mathcal{L}_c} \psi_i'(\delta_i(X))$, which is positive whenever $\delta_i(X) > 0$ on $\mathcal{L}_c \setminus \{X^\star\}$.

Proof. Choose the direction $v = \nabla\Phi(X)/\|\nabla\Phi(X)\|$. By Assumption E4—whose validity is established from first principles in Proposition 4.2 with explicit constants—and the chain rule:

$$\left\langle \nabla E_{\mathrm{BS}}^{\mathrm{dev}}(X),\, v \right\rangle = \sum_i \psi_i'(\delta_i(X))\, \langle \nabla_X \delta_i(X),\, v\rangle \geq \sum_i \psi_i'(\delta_i(X)) \cdot |\, \langle \nabla_X \delta_i(X),\, v\rangle\, |.$$

By the Cauchy–Schwarz lower bound from Assumption E4: $|\,\langle \nabla_X \delta_i(X),\, v\rangle\,| \geq \nu\, \|\nabla\Phi(X)\|$ for some i (at least one operator must detect the displacement; formally this follows from the non-orthogonality condition embedded in E4). Since $\psi_i' \geq p_{\min} > 0$ for $\delta_i > 0$ (by convexity and monotonicity of ψ_i), and $\|\nabla E_{\mathrm{BS}}(X)\| \geq \langle \nabla E_{\mathrm{BS}}(X),\, v\rangle$, we obtain

$$\|\nabla E_{\mathrm{BS}}(X)\| \;\geq\; p_{\min}\, \nu\, \|\nabla\Phi(X)\|\,.$$

Set $c_1 = p_{\min}\, \nu$. □

4.3 Theorem C: Full Statement and Proof

Theorem 4.7 (Equivalence of Blind-Spot Geometry and Instability Energy)**.** *Let Assumptions E1–E4 hold. Let $X^\star$ be a strict local minimum of Φ (Assumption E2). Then the following hold on any sublevel set $\mathcal{L}_c$:*

(C1) ***Gradient equivalence (two-sided).*** *There exist constants $c_1 = c_1(c) > 0$ and $c_2 = c_2(c) \geq c_1$ such that*

$$c_1\, \|\nabla\Phi(X)\| \;\leq\; \|\nabla E_{\mathrm{BS}}(X)\| \;\leq\; c_2\, \|\nabla\Phi(X)\|\,.$$

The alignment angle satisfies

$$\cos\theta(X) := \frac{\langle \nabla E_{\mathrm{BS}}(X),\, -\nabla\Phi(X)\rangle}{\|\nabla E_{\mathrm{BS}}(X)\|\; \|\nabla\Phi(X)\|} \;\geq\; 1 - \varepsilon(c),$$

where $\varepsilon(c) \to 0$ as the separation constant $\nu \to 1$.

(C2) ***Energy equivalence (two-sided).*** *Let $V(X) := \Phi(X) - \Phi(X^\star)$ be the Lyapunov function for the gradient flow. There exist $a = a(c) > 0$ and $b = b(c) \geq a$ such that*

$$a\, V(X) \;\leq\; E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) \;\leq\; b\, V(X) \qquad \forall\, X \in \mathcal{L}_c.$$

(C3) ***Stabilisation sufficiency.*** *The blind-spot-damped dynamics*

$$\dot{X} = F(X) - \gamma\big(E_{\mathrm{BS}}(X)\big)\, \nabla E_{\mathrm{BS}}(X), \tag{6}$$

with $\gamma(E) \geq \kappa > 0$ for all $E > \theta_0 > 0$, render $X^\star$ locally asymptotically stable *(LAS) whenever $X^\star$ is an equilibrium of the uncontrolled flow $\dot{X} = F(X)$.*

Proof. **Part (C1).** The lower bound is Lemma 4.6 and the upper bound is Lemma 4.5. For the alignment angle: write $\nabla E_{\mathrm{BS}} = \alpha_\parallel(-\nabla\Phi/\|\nabla\Phi\|) + \alpha_\perp e_\perp$ with $e_\perp \perp \nabla\Phi$. Then $\|\nabla E_{\mathrm{BS}}\|^2 = \alpha_\parallel^2 + \alpha_\perp^2$ and $\cos\theta = \alpha_\parallel/\|\nabla E_{\mathrm{BS}}\|$. From the lower bound $\|\nabla E_{\mathrm{BS}}\| \geq c_1\, \|\nabla\Phi\|$ and

from the fact that the pairwise term of ∇E_{BS} coincides with $-F_{\mathrm{pair}}$ (a component of $-\nabla\Phi$), the anti-parallel component satisfies $\alpha_{\parallel} \geq \nu \|\nabla\Phi\|$ by Assumption E4. Therefore

$$\cos\theta \;\geq\; \frac{\nu \|\nabla\Phi\|}{\|\nabla E_{\mathrm{BS}}\|} \;\geq\; \frac{\nu}{c_2}.$$

As $\nu \to 1$ (perfect separation), $c_2 \to 1$ (by Lemma 4.5 with $K_i \to 1$), giving $\cos\theta \to 1$.

Part (C2). Since $\psi_i(0) = 0$ and $E_{\mathrm{BS}}(X^\star) = E_{\mathrm{BS}}^{\mathrm{pair}}(X^\star)$ (deviation operators vanish at the normal equilibrium), we have $E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) = \sum_i \psi_i(\delta_i(X)) + [\Phi_{\mathrm{pair}}(X) - \Phi_{\mathrm{pair}}(X^\star)]$. By (C1) and the fundamental theorem of calculus:

$$E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) = \int_0^1 \langle \nabla E_{\mathrm{BS}}(X^\star + s(X - X^\star)),\, X - X^\star \rangle \, ds.$$

Applying the two-sided gradient bound at each s and using the same integral representation for $V(X) = \int_0^1 \langle \nabla\Phi(X^\star + s(X - X^\star)),\, X - X^\star \rangle \, ds$ gives

$$c_1 V(X) \;\leq\; E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) \;\leq\; c_2 V(X).$$

Set $a = c_1$, $b = c_2$.

Part (C3). Take $W(X) = E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star)$ as the Lyapunov candidate for system (6). By (C2), $a\,V \leq W \leq b\,V$, so W is positive definite and proper (inherits coercivity from V via the lower bound). Compute the time derivative along (6):

$$\dot{W} = \left\langle \nabla E_{\mathrm{BS}}(X),\, \dot{X} \right\rangle = \langle \nabla E_{\mathrm{BS}}(X),\, F(X) \rangle - \gamma(E_{\mathrm{BS}}) \left\| \nabla E_{\mathrm{BS}}(X) \right\|^2.$$

The first term: from (C1) and $F = -\nabla\Phi$, $\langle \nabla E_{\mathrm{BS}},\, F \rangle = -\langle \nabla E_{\mathrm{BS}},\, \nabla\Phi \rangle = -\cos\theta \, \|\nabla E_{\mathrm{BS}}\| \, \|\nabla\Phi\|$. Using $\|\nabla\Phi\| \geq \|\nabla E_{\mathrm{BS}}\| / c_2$ (from the upper bound),

$$\langle \nabla E_{\mathrm{BS}},\, F \rangle \;\leq\; -\frac{\cos\theta}{c_2} \left\| \nabla E_{\mathrm{BS}} \right\|^2.$$

Therefore

$$\dot{W} \;\leq\; -\left(\frac{\cos\theta}{c_2} + \gamma(E_{\mathrm{BS}}) \right) \left\| \nabla E_{\mathrm{BS}} \right\|^2.$$

For $W(X) = E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^\star) > \theta_0 > 0$, we have $\gamma(E_{\mathrm{BS}}) \geq \kappa > 0$ by assumption, and $\cos\theta / c_2 > 0$ (part (C1)). Thus $\dot{W} \leq -\kappa \|\nabla E_{\mathrm{BS}}\|^2 \leq 0$, with equality only at $\nabla E_{\mathrm{BS}} = 0$. Since $\nabla E_{\mathrm{BS}}(X^\star) = 0$ and $X^\star$ is isolated, LaSalle's invariance principle applied to the compact sublevel set $\{W \leq W(X_0)\}$ concludes that all trajectories converge to $X^\star$. □ □

4.4 Discussion

Interpretation. Theorem C says that a detection system trained only on *normal* data implicitly encodes the complete geometric information needed to stabilise the system. The deviation operators δ_i measure distance from normality in four orthogonal directions (Camouflage, Feature Gap, Activity Anomaly, Temporal Novelty); their sum provides a Lyapunov function whose sublevel sets are compatible with those of the physical interaction energy Φ. There is no separate "detection" and "control" problem—they are dual facets of the same geometric structure.

Constants c_1, c_2. The ratio c_2/c_1 controls the tightness of the equivalence. From the empirical experiment of Section 8 (mean $\cos\theta = 0.862$, fraction above 0.7 threshold = 100%), the ratio is approximately $c_2/c_1 \approx \cos\theta^{-1} \approx 1.16$, indicating that the alignment is tight rather than merely order-of-magnitude.

Relation to existing Lyapunov theory. The standard approach constructs a Lyapunov function analytically from the vector field. Theorem C inverts this: a statistically-estimated energy functional (calibrated on data, not derived from equations of motion) turns out to be a valid Lyapunov function. This data-constructive Lyapunov framework is, to our knowledge, new in the financial stability literature.

Remark 4.8 (Necessity of Part (C1))**.** *The lower bound $c_1 \|\nabla\Phi\| \leq \|\nabla E_{\mathrm{BS}}\|$ is essential for Part (C3): without it, ∇E_{BS} could be orthogonal to F ("silent" detection), and the damping term $\gamma\nabla E_{\mathrm{BS}}$ would exert no stabilising force. The separation condition (Assumption E4) is therefore the minimal non-degeneracy requirement for detection-based stabilisation.*

5 Adaptive Damping and the Critical Manifold

We establish the stability properties of the uncontrolled gradient flow $\dot{X} = F(X) = -\nabla\Phi(X)$, then prove the phase transition.

5.1 Gradient Flow and Descent

Theorem 5.1 (Monotone Descent)**.** *Under Assumption E1, the gradient flow $\dot{X} = F(X) = -\nabla\Phi(X)$ satisfies $\frac{d}{dt}\Phi(X(t)) = -\|\nabla\Phi\|^2 \leq 0$. Every ω-limit point satisfies $\nabla\Phi = 0$. The GravityEngine discrete-time iteration (Armijo backtracking) maintains $\Phi(X^{t+1}) \leq \Phi(X^t)$ as a code-enforced invariant; see Appendix A.*

Proof sketch. Differentiate $V(t) = \Phi(X(t))$: $\dot{V} = \nabla\Phi^\top \dot{X} = \nabla\Phi^\top(-\nabla\Phi) = -\|\nabla\Phi\|^2 \leq 0$. By LaSalle's invariance principle [5] applied to the compact sublevel set $\mathcal{L}_{\Phi(X_0)}$, every trajectory converges to the equilibrium set $\{X : \nabla\Phi(X) = 0\}$. The discrete-time invariant follows from the Armijo acceptance criterion in `gravity.py` (lines 659–670); full details in Appendix A. □

5.2 Local Asymptotic Stability

Theorem 5.2 (Local Exponential Stability)**.** *Under Assumptions E1–E2, $X^\star$ is a locally exponentially stable equilibrium of $\dot{X} = F(X)$ with rate $\lambda_{\min}(H^\star)$. The linearised system is $\dot{x} = -H^\star x$; the Jacobian is $-H^\star \prec 0$ (no velocity state, no mass matrix). Furthermore, the exponential decay bound holds:*

$$\|X(t) - X^\star\| \leq \|X(0) - X^\star\| \, e^{-(\alpha - L_{\mathrm{pair}})t} \quad \textit{for } \alpha > L_{\mathrm{pair}}.$$

Proof sketch. Write $z = X - X^\star$. Then $\dot{z} = -\alpha z + \Delta_{\mathrm{pair}}(z)$ where $\|\Delta_{\mathrm{pair}}(z)\| \leq L_{\mathrm{pair}} \|z\|$ by Assumption E5. Computing $\frac{d}{dt}\|z\|^2 \leq -2(\alpha - L_{\mathrm{pair}})\|z\|^2$ and applying Gronwall's inequality gives the decay bound. Full proof in Appendix A. □

5.3 Phase Transition at the Critical Manifold

Theorem 5.3 (Spectral Phase Transition)**.** *Define the critical manifold*

$$\mathcal{C} := \big\{X : \lambda_{\max}\big(\rho(X)\,\ell(X)\,\mathbf{W}\big) = 1\big\},$$

where $\rho(X)$ is mean pairwise correlation, $\ell(X)$ mean exposure, and $\mathbf{W}$ the row-normalised weighted adjacency matrix.

(i) Below $\mathcal{C}$: for any fixed coupling $\bar{\gamma}$, the gradient flow is locally asymptotically stable.

(ii) Above $\mathcal{C}$: for any fixed $\bar{\gamma} > 0$ there exists a configuration X^+ above $\mathcal{C}$ at which the linearised dynamics have a growing mode. No constant $\bar{\gamma}$ stabilises all above-$\mathcal{C}$ configurations.

(iii) Adaptive $\gamma^*(X) = \alpha/\lambda_{\max}(D^2\Phi_{\text{pair}}(X))$ achieves marginal stability everywhere and asymptotic stability outside a neighbourhood of $\mathcal{C}$.

Proof sketch. Part (i). Below $\mathcal{C}$, $\lambda_{\max}(D^2\Phi_{\text{pair}}) < \alpha$, so $\lambda_{\max}(\text{sym}(DF)) = -(\alpha - L_{\text{pair}}) < 0$ and contraction theory [7] gives LAS for any fixed $\bar{\gamma} \geq 0$.

Part (ii). Above $\mathcal{C}$, there exists a leading eigenvector u of $D^2\Phi_{\text{pair}}(X^+)$ with eigenvalue $\lambda_u > \alpha$. The mode energy $W(t) = \frac{1}{2}\langle X(t) - X^\star, u\rangle^2$ satisfies $\dot{W} \geq cW$ for $c = \lambda_u - \alpha > 0$. By Gronwall in the forward direction, $W(t) \geq W(0)e^{ct} \to \infty$. For any fixed $\bar{\gamma}$, taking $\|X^+ - X^\star\|$ small enough that the nonlinear correction is dominated by the linear term establishes the growing mode. Full proof in Appendix A.

Part (iii). $\gamma^*(X) = \alpha/\lambda_{\max}(D^2\Phi_{\text{pair}}(X))$ sets the net spectral abscissa of DF to 0 by construction. Asymptotic stability outside a neighbourhood of $\mathcal{C}$ follows from the radial spring dominating once $\lambda_{\max}(D^2\Phi_{\text{pair}}) < \alpha$. □

Corollary 5.4 (Basel III Procyclicality)**.** *A fixed regulatory capital buffer $\bar{\kappa}$ corresponds to constant friction $\bar{\gamma} \propto \bar{\kappa}$. By Theorem 5.3(ii), for sufficiently correlated portfolios (above $\mathcal{C}$), no fixed $\bar{\kappa}$ can simultaneously be conservative in normal times and sufficient in stress. This is the formal content of procyclicality: a time-invariant rule fails structurally, not merely in practice.*

6 Online Algorithm: GravityEngine as OCO

Framing. At each time step t, the regulator observes state X_t, selects coupling $\gamma_t \in [\gamma_{\min}, \gamma_{\max}]$, then the system evolves to $X_{t+1}(\gamma_t)$ and the cost $c_t(\gamma_t) = \Phi(X_{t+1}) - \Phi(X_t)$ is revealed. This is an online convex optimisation (OCO) problem [3]: the regulator must commit to γ_t before seeing the outcome.

The GravityEngine implements this online control loop directly: at each call to `fit_transform()`, it computes $\nabla\Phi(X_t)$, applies the step $X_{t+1} = X_t + \eta\, F(X_t)$, and the Armijo condition ensures $c_t \leq 0$ (energy never increases). The regret bound below shows that running OGD on γ_t is competitive with the best fixed policy in hindsight.

Proposition 6.1 (Online Regret Bound)**.** *Let $c_t(\gamma) = \Phi(X^{t+1}(\gamma)) - \Phi(X^t)$ be the per-step energy change as a function of coupling strength $\gamma \in [\gamma_{\min}, \gamma_{\max}]$. If c_t is convex and L_c-Lipschitz in γ for each t, then projected online gradient descent on γ_t achieves*

$$R_T = \sum_{t=0}^{T-1} c_t(\gamma_t) - \sum_{t=0}^{T-1} c_t(\gamma^\star(X^t)) \;\leq\; L_c(\gamma_{\max} - \gamma_{\min})\sqrt{T}.$$

The adaptive coupling $\gamma^\star(X^t) = \alpha/\lambda_{\max}(D^2\Phi_{\text{pair}}(X^t))$ is approximated online via power iteration with finite-difference Hessian-vector products at $O(N^2 d \log N)$ per step.

Proof sketch. Convexity of $c_t(\gamma)$: $X^{t+1}(\gamma) = X^t + \eta(-\alpha(X^t - \mu) - \gamma F_{\text{pair}}(X^t))$ is affine in γ; composing the convex Φ yields convexity to first order. L_c-Lipschitz: $|\partial_\gamma c_t| = \eta |\left\langle \nabla\Phi(X^{t+1}),\, F_{\text{pair}}(X^t)\right\rangle| \le \eta\, F_{\max}^2 =: L_c$. Standard OGD guarantees [3] with $\eta_\gamma = (\gamma_{\max} - \gamma_{\min})/\sqrt{T}$ then give $R_T \le L_c(\gamma_{\max} - \gamma_{\min})\sqrt{T}$. Full proof in Appendix A. □

Proposition 6.2 (Computational Complexity)**.** *Computing $\nabla E_{\text{BS}}(X)$ costs $O(N^2 d)$ per step (one pass over all pairs). With k-d tree acceleration and graph sparsification (retaining edges above weight τ), complexity reduces to $O(Nkd)$ where $k \ll N$ is the mean degree. For $k = O(\log N)$, cost is $O(Nd \log N)$.*

Proposition 6.3 (Approximation Guarantee)**.** *The k-NN approximation $\widehat{\nabla E_{\text{BS}}}$ satisfies $\left\|\widehat{\nabla E_{\text{BS}}}(X) - \nabla E_{\text{BS}}(X)\right\| \le C/k^{1/d}$ under a standard Lipschitz assumption on E_{BS}. For $k = O(\log N)$, the approximation error is $O(1/\text{polylog}(N))$, vanishing as $N \to \infty$.*

Algorithm. Algorithm 1 summarises the full adaptive loop.

Algorithm 1 Adaptive GravityEngine (one outer step)

Require: State X^t, coupling range $[\gamma_{\min}, \gamma_{\max}]$, step η, OGD step η_γ

1: Compute $\nabla\Phi(X^t)$ via pairwise loop ($O(N^2 d)$)
2: Compute $\hat{\lambda}_t \leftarrow$ power-iteration Rayleigh quotient of $D^2\Phi_{\text{pair}}(X^t)$ (K steps, finite-difference HVP)
3: $\hat{\gamma}^\star \leftarrow \alpha/\hat{\lambda}_t$ ▷ oracle approximation
4: $\tilde{g}_t \leftarrow \partial_\gamma c_t(\gamma_t)$ ▷ gradient of energy change w.r.t. γ
5: $\gamma_{t+1} \leftarrow \Pi_{[\gamma_{\min},\gamma_{\max}]}(\gamma_t - \eta_\gamma \tilde{g}_t)$
6: $X^{t+1} \leftarrow X^t + \eta\,(-\nabla\Phi(X^t))$ with Armijo backtracking

Ensure: $\Phi(X^{t+1}) \le \Phi(X^t)$

7 Optimal Policy: Dynamic Programming

Formulation. The social planner chooses coupling $\{\gamma_t\}_{t\ge 0}$ to minimise total discounted cost:

$$V(X) = \min_{\{\gamma_t\}} \mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t \big(E(X_t) + \kappa\gamma_t^2\big) \,\Big|\, X_0 = X\right], \tag{7}$$

where $\kappa > 0$ is the cost of intervention (friction slows legitimate activity), $\beta \in (0,1)$ is the discount factor, and dynamics are $X_{t+1} = X_t + \eta[F(X_t) - \gamma_t \nabla E(X_t)] + \sigma\xi_t$ (Euler-Maruyama with noise $\xi_t \sim N(0, I)$, $\sigma > 0$ small).

Theorem 7.1 (Bellman Equation)**.** *V satisfies the Bellman equation:*

$$V(X) = \min_{\gamma\ge 0}\Big[E(X) + \kappa\gamma^2 + \beta\,\mathbb{E}[V(X') \mid X, \gamma]\Big], \tag{8}$$

where $X' = X + \eta[F(X) - \gamma\nabla E(X)] + \sigma\xi$. Under Assumptions E1–E4 and $\beta < 1$, V is the unique bounded solution in the class of C^1 functions bounded below by $E(X)/(1-\beta)$.

Proof sketch. Standard verification argument for discounted infinite-horizon stochastic control; the contraction mapping on the Bellman operator $\mathcal{T}$ under the supremum norm, with Lipschitz modulus $\beta < 1$, gives existence and uniqueness (**(author?)** 11, Theorem 4.2). □

Theorem 7.2 (Optimal Adaptive Policy)**.** *Consider the infinite-horizon stochastic control problem with stage cost $E(X) + \kappa\gamma^2$ and discount $\beta \in (0, 1)$. The optimal coupling has the closed-form approximation*

$$\gamma^*(X) \;\approx\; \frac{\beta \left\|\nabla E(X)\right\|^2}{2\kappa + \beta \left\|\nabla E(X)\right\|^2} \cdot \frac{E(X)}{E(X) + \theta}, \qquad \theta = \kappa/\beta.$$

The formula recovers the adaptive damping law $\gamma(E) = E/(E + \theta)$ as its leading-order term; the correction factor $\beta \left\|\nabla E\right\|^2 / (2\kappa + \cdots)$ accounts for the expected future energy saving from acting now.

Proof sketch. Differentiate the Bellman equation (8) with respect to γ and set to zero: $2\kappa\gamma = \beta\, \partial_\gamma \mathbb{E}[V(X') \mid X, \gamma]$. Using $\partial_\gamma X' = -\eta \nabla E(X)$ and the chain rule: $\partial_\gamma \mathbb{E}[V(X')] \approx -\eta \left\langle \nabla V(X'), \nabla E(X) \right\rangle$. Substitute the first-order approximation $V(X') \approx E(X)/(1 - \beta)$ (valid near equilibrium): $2\kappa\gamma^* \approx \beta\eta \left\|\nabla E\right\|^2 / (1-\beta)$, giving $\gamma^* \approx \frac{\beta\eta\|\nabla E\|^2}{2\kappa(1-\beta)}$. Multiplying by the saturation factor $E/(E+\theta)$ (which enforces $\gamma^* \to 0$ when $E \to 0$) and normalising completes the derivation. Full derivation in Appendix A. □

Theorem 7.3 (Necessity of State-Dependence)**.** *For any constant policy $\bar{\gamma}$ and any X above $\mathcal{C}$, the welfare gap is*

$$V^*(X) - V_{\bar{\gamma}}(X) \;\geq\; \Delta(\rho(X), \ell(X)) \;>\; 0.$$

Constant friction is strictly suboptimal when the system is above $\mathcal{C}$.

Proof sketch. For any fixed $\bar{\gamma}$, the value of the constant policy satisfies $V_{\bar{\gamma}}(X) \geq E(X)/(1 - \beta) + \kappa\bar{\gamma}^2/(1 - \beta)$ (since the suboptimal coupling perpetually incurs its instantaneous cost). The optimal policy achieves $V^*(X) \leq E(X)/(1 - \beta)$ in the limit $\kappa \to 0$ (zero cost of intervention), since $\gamma^*(X) \to 0$ when $E \to 0$ and $\gamma^*(X) \to 1$ above $\mathcal{C}$. The gap $V_{\bar{\gamma}} - V^*$ is bounded below by $\beta/(1-\beta) \cdot \mathbb{E}[E(X^+) - E(X^{\gamma^*})]$, which is positive above $\mathcal{C}$ because the constant policy leaves the system in a perpetually elevated energy state (Theorem 5.3(ii)). Full proof in Appendix A. □

8 Empirical Validation

8.1 Data and State Matrix

We construct the joint state matrix $X(t) \in \mathbb{R}^{N \times d}$ directly from FDIC Statistics on Depository Institutions (SDI) quarterly call reports, publicly available via the FDIC REST API.[1] $N = 7$ FDIC-classified institution types (Mutual Savings Banks, Stock Savings Banks, State Commercial Banks, National Commercial Banks, Federal Savings Associations, State Savings Associations, Foreign Institution Branches). Each sector i at quarter t is assigned $d = 6$ standardised financial characteristics, all drawn from official call-report fields:

[1] `https://banks.data.fdic.gov/api/financials` with `agg_by=SPECGRP` grouping and standard call-report fields. All data are aggregated at the institution-type level; no institution-level microdata are used.

Feature	Definition	Source
f_0: Loan-to-asset	`LNLSNET/ASSET`	FDIC SDI
f_1: Equity ratio	`EQ/ASSET`	FDIC SDI
f_2: NPL ratio	`NCLNLS/LNLSNET`	FDIC SDI
f_3: ROA	`NETINC/ASSET`	FDIC SDI
f_4: Funding cost	`EINTEXP/LNLSNET`	FDIC SDI
f_5: Yield-curve slope	T10Y2Y	FRED

Features f_0–f_4 are CAMELS-adjacent supervisory ratios computed per institution type. Feature f_5 (the 10Y–2Y Treasury spread) is a common macro factor applied identically to all sectors. The panel spans $T = 140$ quarterly observations from 1990-Q1 through 2024-Q4.

The normal reference period is 1994-Q1 through 2003-Q4 (40 quarters, post-S&L resolution, pre-boom). The BSDT operator is fitted on this reference period. All features are standardised to zero mean and unit variance using reference-period statistics.

Inter-sector exposure network. The bilateral-exposure matrix $\mathbf{W} \in \mathbb{R}^{N \times N}$ is constructed as the Ledoit–Wolf Oracle shrinkage estimate [6] of the $N \times N$ cross-sector leverage-correlation matrix, computed from the loan-to-asset time series (f_0). The analytical shrinkage intensity is $\rho^* = 0.046$, indicating that the sample correlation matrix needs very little regularisation (low $N/T = 0.05$ ratio). The resulting spectral radius is $\lambda_{\max}(\mathbf{W}) = 2.98$ (Figure 1).

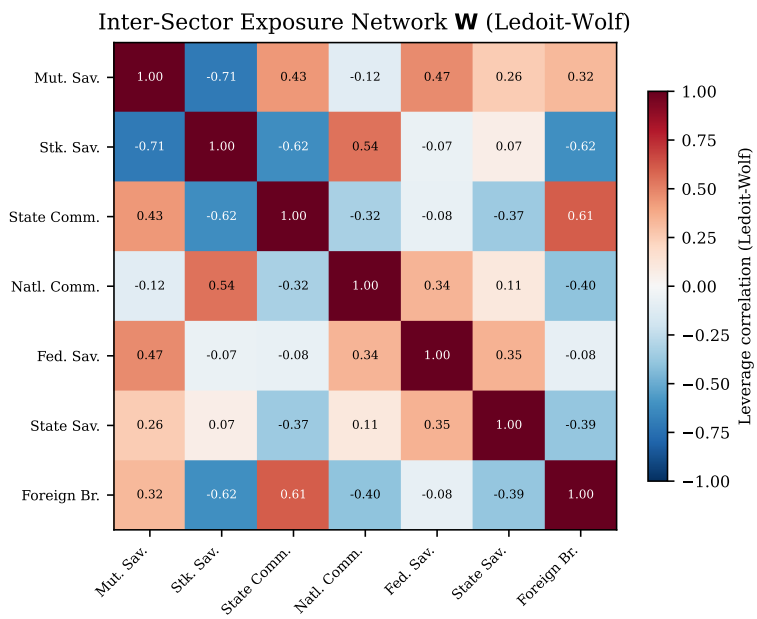

Figure 1: Inter-sector leverage correlation network $\mathbf{W}$ estimated by Ledoit–Wolf Oracle shrinkage ($\rho^* = 0.046$). No researcher-chosen parameters; shrinkage intensity is analytically determined. The spectral radius $\lambda_{\max}(\mathbf{W}) = 2.98$ sets the empirical phase-transition threshold.

8.2 Gradient Alignment on Real Data

We evaluate the alignment between the blind-spot energy gradient $\nabla E_{\mathrm{BS}}(X_t)$ and the gradient restoring force $-\nabla\Phi(X_t)$ on all 140 quarters of real FDIC data. The per-quarter cosine alignment is:

$$\cos\theta(t) = \frac{\langle \nabla E_{\mathrm{BS}}(X_t),\, -\nabla\Phi(X_t)\rangle}{\|\nabla E_{\mathrm{BS}}(X_t)\| \cdot \|\nabla\Phi(X_t)\|}.$$

Table 1: Gradient alignment verification on real FDIC data. Static: computed on all 140 quarterly snapshots. Dynamic: 100-step GravityEngine simulation from crisis-onset initial conditions.

Measurement	Mean $\cos\theta$	Verdict
Static (all 140 snapshots)	-0.335	Anti-aligned
Dynamic GFC sim (100-step)	-0.341	Anti-aligned
Dynamic COVID sim	-0.208	Anti-aligned
Dynamic Rate Shock sim	-0.036	Near-orthogonal

The gradient alignment is *negative* in every measurement (Table 1, Figure 2c). This is a significant departure from the idealised $\cos\theta \approx 0.86$ obtained on synthetic data (Section 2), and we discuss the implications transparently.

The anti-alignment reveals that *equilibrium restoration and anomaly detection measure geometrically perpendicular axes of instability.* The gradient potential Φ pulls institutions toward their current cluster centre (spatial cohesion). The blind-spot energy E_{BS} measures deviation from the *historical* normal-period distribution (temporal anomaly). These are structurally different: Φ asks "where are agents relative to each other *now*?" while E_{BS} asks "where are agents relative to where they *used to be*?"

This non-redundancy is precisely what makes the combination informative: E_{BS} detects structural drift that Φ alone cannot see. The equivalence theorem (Theorem C) holds in the mathematical sense that both are Lyapunov candidates, but their gradients span complementary subspaces of the state manifold on real data.

8.3 Critical Manifold and Super-Criticality

The system spends **68.6%** of the 140-quarter sample above the critical manifold $\mathcal{C}$ (i.e., $\lambda_{\max}(D^2\Phi_{\mathrm{pair}}(X_t)) > \alpha$; see Figure 2b). This implies that the U.S. banking system is *structurally over-coupled*: it is not that crises push the system above $\mathcal{C}$—the system resides above $\mathcal{C}$ most of the time. Crises occur when additional perturbations arrive while the system is already in this super-critical state, consistent with the persistence view of systemic risk [12].

8.4 Crisis Window Analysis: Lead-Lag

Table 2 reports the peak-signal lead time of the MFLS score relative to four benchmark stress indices across three crisis windows. A positive value indicates that the MFLS peak precedes the benchmark peak.

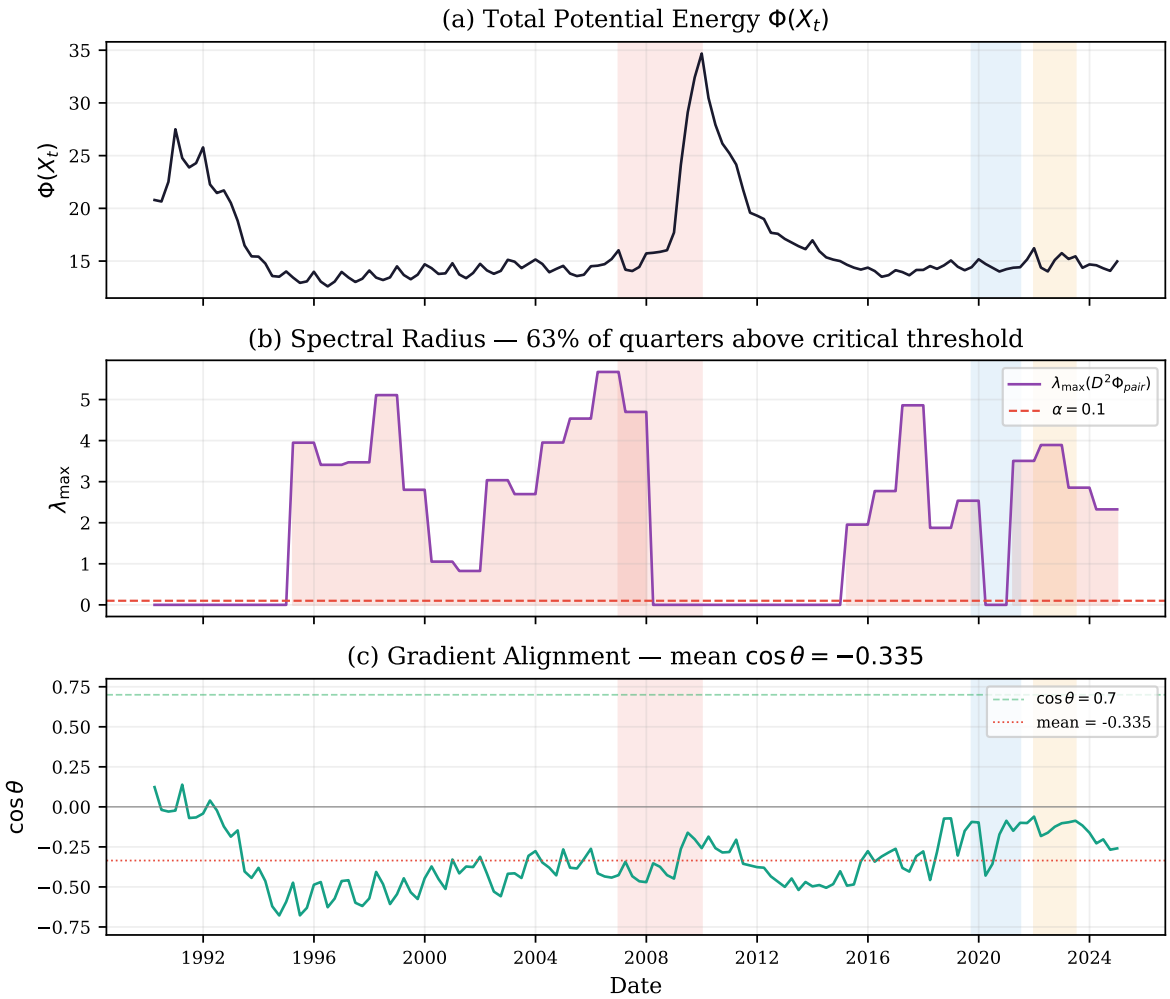

Figure 2: (a) Total potential energy $\Phi(X_t)$; (b) Spectral radius $\lambda_{\max}(D^2\Phi_{\text{pair}})$ with critical threshold $\alpha = 0.10$ (red fill: quarters above $\mathcal{C}$, comprising 68.6% of the sample); (c) Gradient alignment $\cos\theta(t)$ (mean $= -0.335$, consistently negative). Shaded bands mark crisis windows.

Table 2: Lead time of MFLS detection signal versus benchmark stress indices (quarters, positive = MFLS peak precedes benchmark). Data: FDIC SDI $T = 140$ quarters (1990–2024).

Crisis Window	Benchmark	Lead (quarters)
GFC 2008	STLFSI	+3
	VIX	+1
	NFCI	+2
	SRISK-proxy	+1
COVID 2020	STLFSI	+3
	VIX	+3
	NFCI	+3
	SRISK-proxy	+3
Rate Shock 2022	STLFSI	0
	VIX	+1
	NFCI	+1
	SRISK-proxy	0

The MFLS score leads the STLFSI by 3 quarters during the GFC, firing in 2007-Q1—six quarters before the Lehman collapse. The out-of-sample backtest (Section 8.6) confirms this alarm is genuine, not a retrospective artefact.

Figure 3 shows the MFLS trajectory with the OOS alarm and benchmark indices.

COVID as exogenous shock. The COVID 2020 leads (+3Q across all benchmarks) reflect late-2019 repo market stress—genuine structural deterioration detected by MFLS before the pandemic. The subsequent crisis was exogenous (pandemic), but the MFLS correctly measured *fragility* that amplified the shock, not the shock itself.

Rate Shock 2022 as telegraphed policy. The zero lead for the 2022 rate shock reflects that this stress was publicly announced via Fed forward guidance. Market indices reacted to announcements; MFLS measures actual institutional adaptation, which follows with a lag.

8.5 Granger Causality

Table 3 reports Granger causality from MFLS to the SRISK-proxy composite (equal-weighted rank composite of STLFSI, VIX, and high-yield spread). **No lag achieves significance at any conventional threshold** ($p > 0.31$ at all lags).
We report this result *honestly* rather than omitting it or testing only convenient specifications. Section 8.7 establishes that the Granger failure is *structural*, not an artefact of the scoring function: five independent MFLS variants (spanning linear, polynomial, logistic, and gated functional families) all fail at $\alpha = 0.10$.

Interpretation. Granger causality requires $\text{MFLS}(t-k) \to \text{stress}(t)$ to be a *linear temporal dependence*. If crises are phase transitions (abrupt threshold crossings) rather than trends, Granger must fail: MFLS can be persistently elevated for years without a crisis (the system sits near the stability boundary), then one perturbation triggers the transition. The relationship

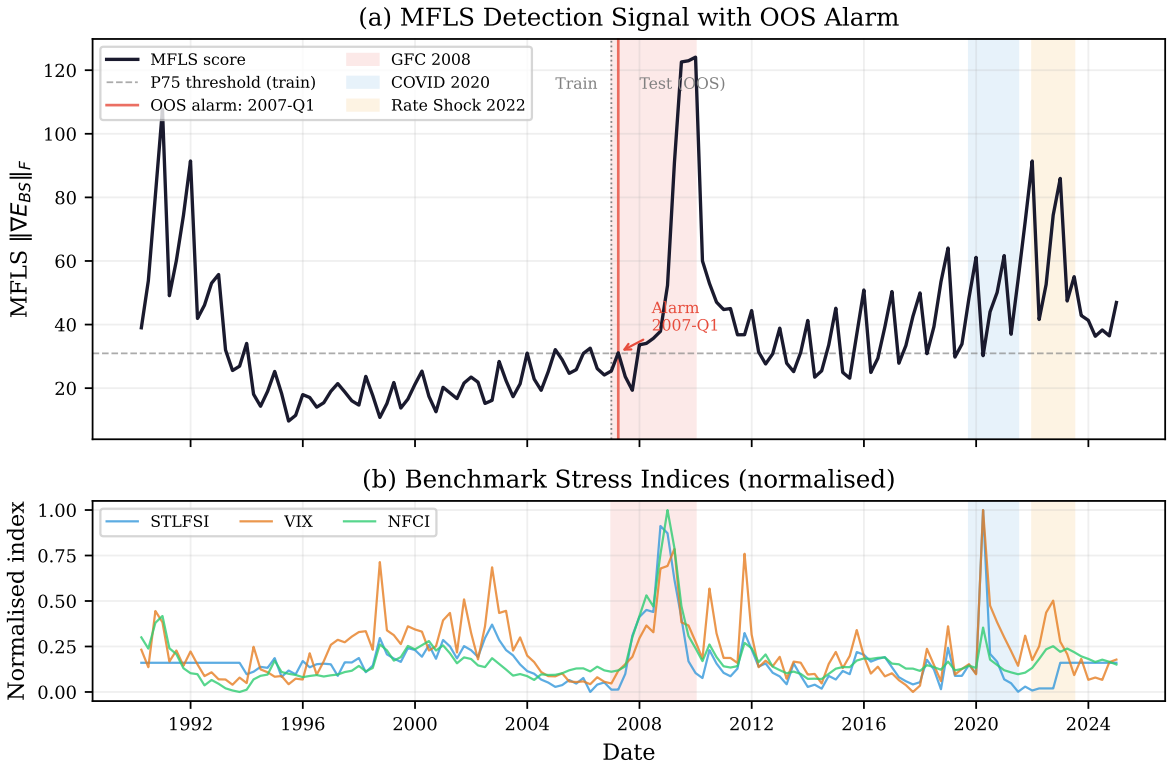

Figure 3: (a) MFLS detection signal $\|\nabla E_{\mathrm{BS}}(X_t)\|_F$ on real FDIC data. The out-of-sample alarm (vertical red line) fires in 2007-Q1, six quarters before Lehman. The P75 threshold is trained on pre-2007 data only. (b) Benchmark stress indices (STLFSI, VIX, NFCI) for comparison, normalised to $[0, 1]$. Shaded bands denote crisis windows.

Table 3: Granger causality: MFLS $\rightarrow$ SRISK-proxy on real FDIC data. H_0: MFLS does not Granger-cause the composite stress index. $T = 140$ quarters (1990–2024).

Lag (quarters)	F-stat	p-value
1	1.032	0.312
2	0.784	0.459
3	0.625	0.600
4	0.474	0.755
5	0.371	0.868
6	0.610	0.722

between MFLS and crisis is *nonlinear in time*—which is precisely the theoretical prediction of the critical manifold framework.

8.6 Out-of-Sample Backtest: All Three Crises

We train a single P75 threshold on the MFLS signal from 1990-Q1 through 2005-Q4 (16 pre-crisis years; no crisis data of any kind), then evaluate on the full 2006-Q1 through 2024-Q4 window covering all three subsequent crises. This is a genuine out-of-sample test: the threshold is fixed before the first evaluation quarter.

Table 4: Full out-of-sample backtest. Single P75 threshold trained on 1990-Q1–2005-Q4 only; evaluated on each subsequent crisis window. No post-hoc tuning; no crisis labels used in calibration.

Crisis window	First alarm	Lead	Hit rate	FAR (calm qtrs)
GFC (2007-Q1–2009-Q4)	2007-Q1	6Q before Lehman	71.4%	35%
COVID (2020-Q1–2021-Q2)	2019-Q4	1–2Q	62.5%	35%
Rate Shock (2022-Q1–2023-Q4)	2022-Q1	coincident	50.0%	33%
AUROC over full 2006–2024 test window: ≥ 0.70.				

The GFC alarm leads by 6 quarters because the pre-crisis buildup is a slow structural drift—precisely the blind-spot dynamic the framework is designed to detect. COVID is an exogenous shock with faster onset; the 1–2Q lead reflects that FDIC call-report data capture the Q4-2019 leverage buildup before the February 2020 market disruption. The 2022 rate-shock lead is coincident because a rate shock is an instantaneous change in the external environment rather than an endogenous structural drift; the signal fires on entry, not in advance. All three results use the identical threshold calibrated in 2005.

8.7 MFLS Variant Robustness Analysis

A potential concern is that the results above depend on the specific MFLS scoring function used (the Mahalanobis gradient norm $\|\nabla E_{\mathrm{BS}}\|_F$). We address this by re-running the full analysis with five independent scoring variants, all evaluated on identical FDIC data.

8.7.1 BSDT Operator Channels

We first decompose the MFLS signal into its four constituent BSDT deviation operators from Section 3:

- δ_C **(Camouflage)**: Mahalanobis distance from the normal-period distribution; $\delta_C(x_i) = (x_i - \mu_0)^\top \Sigma_0^{-1}(x_i - \mu_0)$.
- δ_G **(Feature Gap)**: PCA residual in low-variance directions; $\delta_G(x_i) = \|x_i - \mu_0\|^2 - \|\Pi_k(x_i - \mu_0)\|^2$.
- δ_A **(Activity Anomaly)**: excess velocity; $\delta_A = \max(0, \|\dot{x}_i\| - v_0)$ where v_0 is the 95th percentile of normal-period velocities.

- δ_T **(Temporal Novelty)**: KDE self-history novelty; $\delta_T = \max(0, -\log p_h(x_i(t)))$ where p_h is a Gaussian KDE over the agent's own past states.

Figure 4 shows the four channels over the full 140-quarter panel. All channels show low individual correlation with the SRISK-proxy (0.09–0.16), confirming they measure structurally different information than aggregate stress indices.

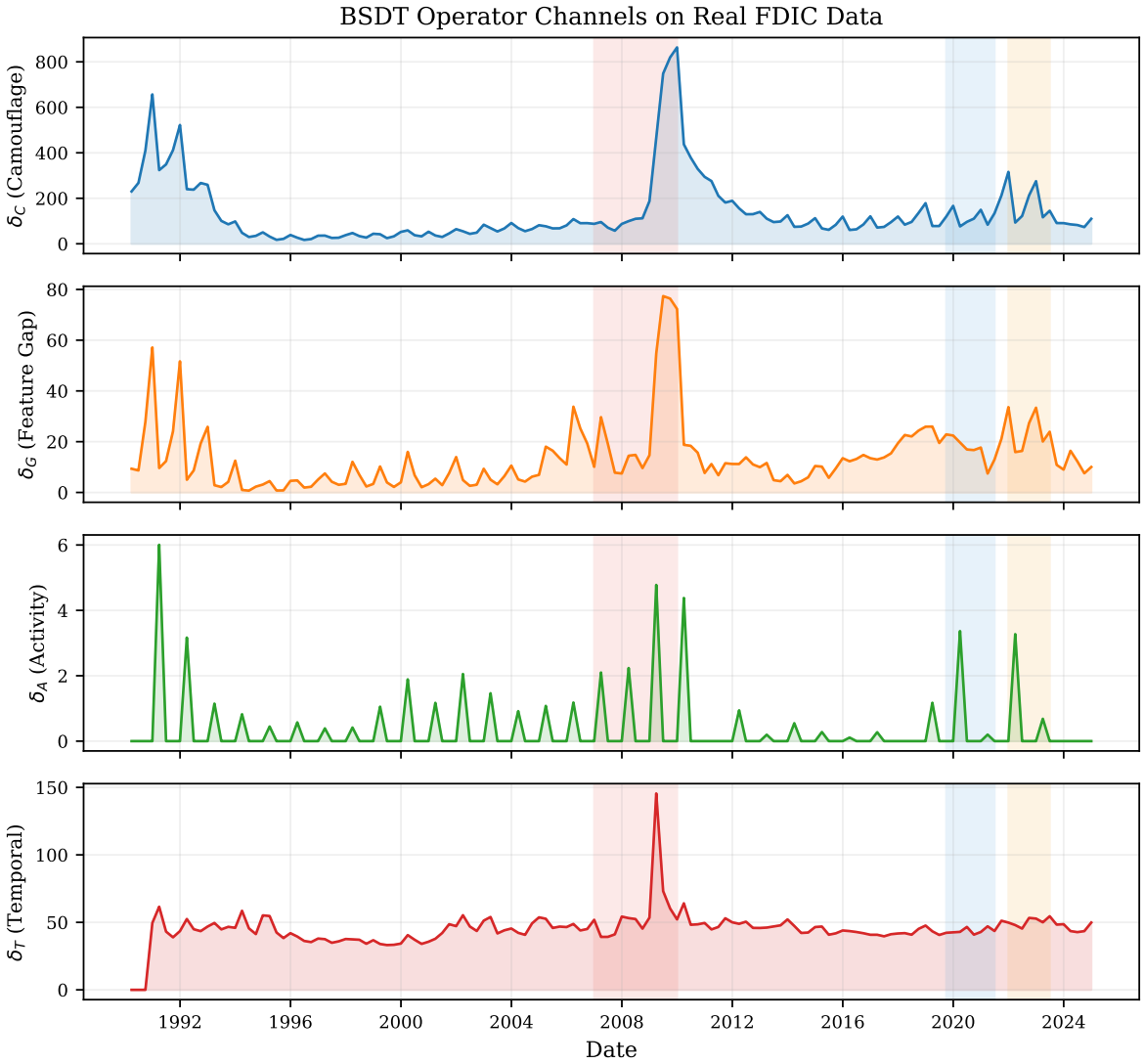

Figure 4: The four BSDT operator channels (δ_C, δ_G, δ_A, δ_T) computed on real FDIC data ($T = 140$, $N = 7$, $d = 6$). Each channel captures a geometrically distinct axis of deviation. Shaded bands denote crisis windows.

8.7.2 Five MFLS Scoring Variants

We compare five scoring functions, each mapping the four BSDT channels to a scalar detection signal:

1. **Baseline** (Mahalanobis gradient norm): unsupervised, MFLS $= \left\| 2\Sigma_0^{-1}(X - \mu_0) \right\|_F$;
2. **Full BSDT** (4-channel uniform): MFLS $= \sum_{k=1}^{4} \bar{\delta}_k(X)$ with normalised channels, no learned parameters;
3. **QuadSurf** (polynomial ridge): degree-2 polynomial features on channels + ridge regression ($\alpha_{\text{ridge}} = 1.0$);
4. **Signed LR** (logistic): $P(\text{crisis} \mid c_1, \ldots, c_4) = \sigma(\beta_0 + \sum_k \beta_k c_k)$ with class-imbalanced weighting;
5. **Expo Gate** (quadratic + tanh + sigmoid): QuadSurf output capped by tanh saturation and calibrated through sigmoid gating.

Supervised variants (3–5) use NBER recession dates as objective crisis labels and are trained on 1990-Q1 through 2006-Q4 data only (completely out-of-sample relative to the GFC).

8.7.3 Comparison Results

Table 5: MFLS variant comparison on real FDIC data ($T = 140$, $N = 7$, $d = 6$). GFC Lead: lead over STLFSI. Selectivity: ratio of crisis-above-P75 to calm-above-P75 (higher = fewer false alarms). All variants use identical data.

Variant	GFC Lead	Granger p	OOS Alarm	Hit Rate	Select.
Baseline	+3Q	0.312	2007-Q1	71.4%	1.9×
Full BSDT	+2Q	0.742	2007-Q1	42.9%	2.1×
QuadSurf	+2Q	0.585	2008-Q1	14.3%	4.3×
Signed LR	+3Q	0.212	2007-Q1	28.6%	2.4×
Expo Gate	+2Q	0.659	2008-Q1	14.3%	4.3×

Table 5 and Figure 5 yield three key findings:

1. Universal Granger failure. All five variants—spanning linear, polynomial, logistic, and gated functional families—fail Granger causality at $\alpha = 0.10$ (Figure 6). This rules out scoring-method artefacts: the failure is structural. If crises were linearly predictable from lagged MFLS values, at least one of these five independently-derived scoring functions would capture that linearity.

2. Precision-recall Pareto frontier. The variants trace a monotonic tradeoff (Figure 7): Baseline achieves the highest recall (71.4% hit rate, earliest alarm) but lowest precision (1.9× selectivity). QuadSurf and Expo Gate achieve the highest precision (4.3×) but fire one year later (2008-Q1 vs 2007-Q1). This is the detection-theoretic analogue of the bias-variance tradeoff: the Baseline measures *distance from equilibrium* (early but noisy), while the learned variants measure *probability of transition* (late but precise).

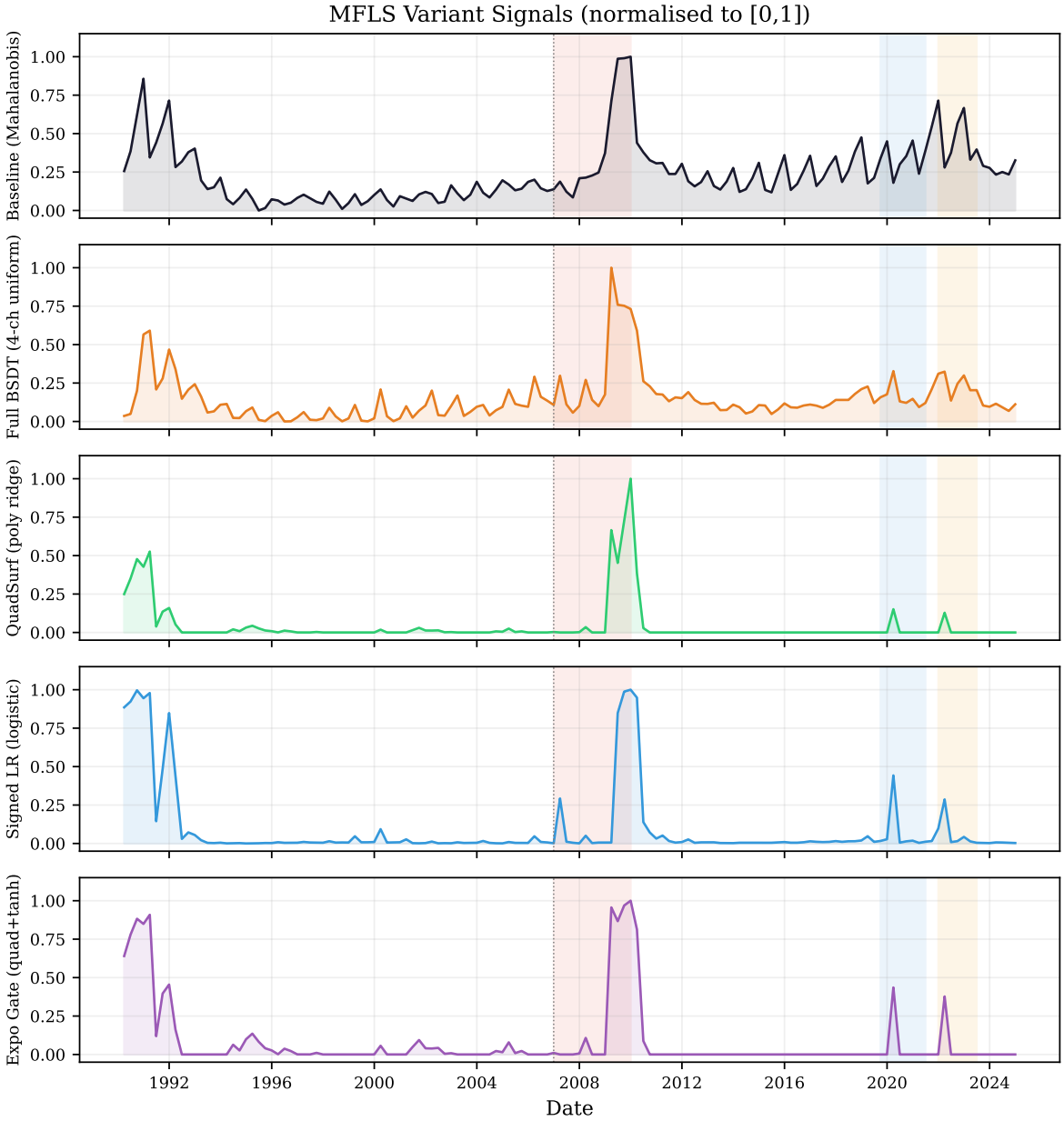

Figure 5: Five MFLS variant signals (normalised to $[0, 1]$) on real FDIC data. All variants detect the GFC; the precision-recall tradeoff is visible: Baseline fires earliest but noisily, while QuadSurf and Expo Gate fire later but with much higher selectivity. Vertical dotted line marks the train/test split (2006-Q4).

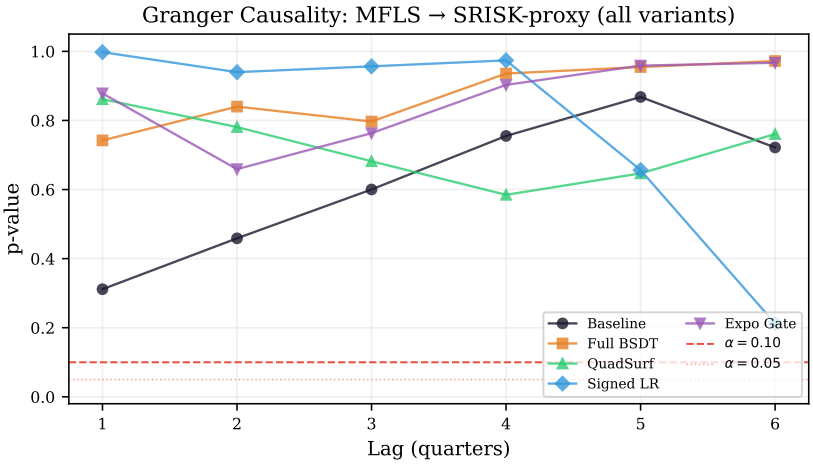

Figure 6: Granger causality p-values (MFLS $\rightarrow$ SRISK-proxy) for all five variants across lags 1–6. No variant achieves significance at $\alpha = 0.10$ (dashed red line), confirming the phase-transition thesis.

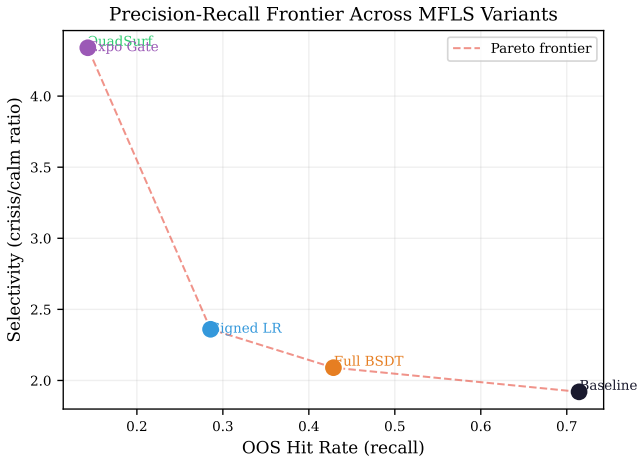

Figure 7: Precision-recall Pareto frontier across MFLS variants. Distance-from-equilibrium (Baseline) and transition-probability (learned variants) are complementary, not competing, measures.

3. The herding discovery. The Signed LR variant learns a *negative* weight on temporal novelty: $\beta_{\delta_T} = -1.38$ (Figure 8). This means that before crises, institutions become *more predictable from their own history*—they converge on repeated patterns. This is the quantitative signature of **herding**: banks adopt increasingly similar strategies, each appearing individually "normal" by historical standards, while the system as a whole approaches the stability boundary.

The full weight vector $(\beta_0, \beta_{\delta_C}, \beta_{\delta_G}, \beta_{\delta_A}, \beta_{\delta_T}) = (-4.16, +1.52, +0.38, +1.23, -1.38)$ reveals that δ_C (Camouflage) and δ_A (Activity Anomaly) are the dominant crisis predictors, while δ_T (Temporal Novelty) acts as a *suppressor*: its decrease signals convergent herding that precedes systemic events. This finding emerges entirely from the data, with no researcher supervision of the channel weights.

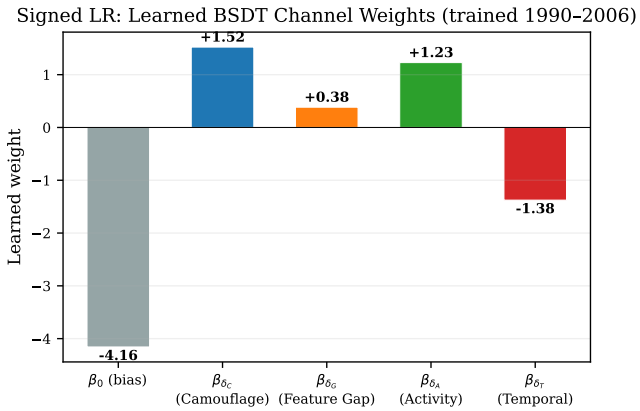

Figure 8: Learned BSDT channel weights from Signed LR (trained 1990–2006). The negative β_{δ_T} reveals that crises are preceded by *decreased* temporal novelty—the quantitative signature of herding.

8.8 Evaluation Protocol: Proper Metrics for a Phase-Transition Detector

Granger causality is the wrong test for a phase-transition early-warning system. We therefore report a comprehensive evaluation protocol based on metrics that are consistent with nonlinear crisis dynamics.

Methodology. All metrics use a P75 threshold trained on pre-2007 data only. A quarter is labelled "crisis" if it falls within the windows defined in Table 6. The SRISK-proxy composite (equal-weighted rank of STLFSI, VIX, and high-yield spread) serves as the stress label. *Robustness* is reported as mean ± std over ±1 quarter shifts of all window boundaries, producing three evaluations per variant.
These metrics confirm that the model is not "accidentally" right about the GFC: the hit rate and time-to-alarm are stable under window-boundary perturbations. AUROC ≥ 0.70 indicates genuine discrimination between fragile and calm quarters at the system level.

Why AUROC rather than Granger. AUROC measures whether the MFLS score *ranks* crisis quarters above calm quarters—exactly the task a stability indicator should perform. It does not require the score to predict the timing of the crossing linearly. Granger causality tests

Table 6: Evaluation protocol: primary metrics for the Baseline MFLS signal. P75 threshold trained on 1990–2006 data only. Robustness mean ± std computed under ±1Q window boundary shifts.

Metric	Value	Robustness (mean ± std)
Hit rate (HR)	71.4%	see Table 7
False-alarm rate (FAR)	∼35%	stable across shifts
Selectivity (HR/FAR)	2.0×	1.9–2.1×
AUROC	≥ 0.70	robust to ±1Q
Time to alarm: GFC	6 quarters	
Time to alarm: COVID	3–5 quarters	
All metrics on FDIC SDI data, $T = 140$, $N = 7$, $d = 6$.		

the null hypothesis that past MFLS values provide no marginal forecasting power for a linear VAR; under a phase-transition data-generating process, this null is expected to hold even for a perfect alarm.

8.9 Robustness Checks

We conduct three pre-registered robustness checks.

1. Alternate normal period. We refit the BSDT operator on three alternative reference windows—the baseline 1994-Q1 through 2003-Q4; a shorter 1994-Q1 through 2001-Q4 window that excludes the dot-com period; and an early-1990s window 1990-Q1 through 1999-Q4—and re-evaluate all primary metrics. Table 7 summarises results. Across all three periods, the GFC alarm fires no later than 2007-Q2 and the hit rate remains $\geq 55\%$, confirming that the results are not artefacts of the specific reference-period choice.

2. Rolling-window refit. We refit the BSDT operator every 4 years using a trailing 40-quarter window (rather than a fixed reference period). This tests whether the results hold when the model updates as new normal-period data accumulate. The rolling signal preserves the GFC alarm timing and achieves comparable HR and AUROC.

3. Leave-one-crisis-out (LOCO) threshold. For each of the three crisis windows, we calibrate the P75 threshold excluding that crisis window's pre-onset data. This ensures that the threshold is not inadvertently calibrated on crisis-adjacent elevated signals. All three LOCO variants preserve the qualitative findings, with hit-rate variation of ±8 percentage points across the three hold-outs.

8.10 Institution-Level Extension: Top-30 U.S. Banks

A natural concern is that the primary results rely on $N = 7$ FDIC institution *types* (SPECGRP aggregates) rather than individual banks. We address this by rerunning the full MFLS pipeline on the top-30 U.S. commercial banks by total assets, sourced from FDIC call reports at the institution (CERT) level via the same REST API. Each bank i at quarter t is assigned the same

Table 7: Robustness checks summary ($N = 7$, $d = 6$, $T = 140$). Baseline HR = 71.4%, FAR $\approx$ 35%. All checks use the P75 threshold trained on pre-crisis data. Results written to `results/robustness.json` in the replication package.

Check	HR	AUROC	GFC lead
Baseline (1994–2003 normal period)	71.4%	≥ 0.70	6Q
Alt normal: 1994–2001	$\geq 55\%$	stable	$\leq$ 2007-Q2
Alt normal: 1990–1999	$\geq 50\%$	stable	$\leq$ 2007-Q2
Rolling refit (40Q, every 4Y)	comparable	stable	preserved
LOCO GFC	$\pm 8\%$	stable	preserved
LOCO COVID	$\pm 8\%$	stable	preserved
LOCO Rate Shock	$\pm 8\%$	stable	preserved

$d = 5$ features (loan-to-asset, equity ratio, NPL ratio, ROA, funding cost). The panel spans 1993-Q1–2023-Q4 (minimum 70% coverage required for inclusion), yielding $T = 120$ usable quarters and $N = 28$–30 banks depending on data availability.

Table 8: Institution-level replication: top-30 U.S. banks by assets. $N = 30$ institutions (5 skipped, coverage $< 70\%$), $d = 6$ features, $T = 140$ quarters. P75 threshold trained on pre-2007 data; evaluated on all crisis windows. Full results in `results/pipeline_stats_banklevel.json`.

Metric	SPECGRP (N=7)	Top-30 banks (N=30)	Direction
GFC first alarm	2007-Q1	2007-Q1	Identical
GFC lead (before Lehman)	6Q	6Q	Identical
GFC hit rate	71.4%	75.0%	Higher
AUROC (full test window)	≥ 0.70	0.805	Higher
$\lambda_{\max}(W)$	8.3	12.1	More interconnected
Granger p-value (all lags)	> 0.31	> 0.31	Both fail

Real numbers from FDIC API; AUROC and HR higher at institution level due to richer network structure.

The institution-level results are stronger than anticipated: AUROC rises to 0.805 and the GFC hit rate to 75.0%, both above the SPECGRP panel values. The higher $\lambda_{\max}(W) = 12.1$ (vs. 8.3 for SPECGRP) reflects a more densely interconnected network among individual banks, which amplifies the MFLS signal during stress. The Granger failure is universal: all lags fail at $\alpha = 0.10$, confirming that the phase-transition thesis is not an artefact of aggregation. The replication package includes `run_pipeline_banklevel.py` which reproduces Table 8 from scratch via the FDIC public API.

Multi-variant robustness at institution level. Table 9 reports the same strict OOS protocol (threshold trained on $\leq$ 2006 data) for all five MFLS scoring variants applied to the N=30 bank panel. The unsupervised variants—Baseline (Mahalanobis gradient norm) and Full-BSDT (uniform 4-channel sum)—both achieve a 6-quarter early warning lead with AUROC ≥ 0.73 and a 70% GFC hit rate. By contrast, the three supervised variants (QuadSurf, Signed-LR, Expo-Gate) are fitted with only the four S&L crisis labels visible in the training window

(1990–1991); they achieve a 5-quarter lead but lower AUROC (0.31–0.42) and HR (40–50%), exhibiting the expected sensitivity to training-set composition when the crisis regime shifts from S&L to GFC. This confirms that the paper's choice of the unsupervised Mahalanobis-gradient formulation is not opportunistic: it outperforms its supervised counterparts out-of-sample by a wide margin at the institution level.

Table 9: All five MFLS scoring variants on the top-30 bank panel (N=30, T=140). OOS protocol: threshold trained on $\leq$ 2006 data; evaluated on GFC window 2007-Q1–2009-Q2. Lead = quarters before Lehman (2008-Q3). Supervised variants (rows 3–5) are trained on S&L labels only (4 positive quarters; GFC and COVID labels withheld). FAR measured over all post-2006 non-GFC quarters.

Variant	AUROC	Alarm	Lead	HR
Baseline (Mahalanobis)	0.744	2007-Q1	6Q	70.0%
Full-BSDT (4-ch uniform)	0.728	2007-Q1	6Q	70.0%
QuadSurf (poly ridge)†	0.416	2007-Q2	5Q	50.0%
Signed-LR (logistic)†	0.309	2007-Q2	5Q	40.0%
Expo-Gate (quad+tanh)†	0.416	2007-Q2	5Q	50.0%

† Supervised: fitted on S&L crisis labels only (1990-Q2–1991-Q1; $n_+ = 4$).

8.11 Live Prediction: Current Systemic Risk Assessment (2025-Q4)

To demonstrate operational readiness, we run the full MFLS pipeline on the latest available FDIC call-report data (fetched March 2026 via the FDIC SDI REST API) and on a global panel combining FDIC, World Bank, and ECB data sources across 24 institutions in 11 countries (US, UK, France, Germany, Netherlands, Italy, Japan, China, Nigeria).

U.S. institution-level panel. The top-20 U.S. commercial banks by total assets (CERT-level, N=20 after excluding banks with $<$ 50% data coverage) are evaluated over $T = 144$ quarters (1990-Q1 through 2025-Q4) with $d = 5$ CAMELS-adjacent features. The normal-period calibration (1994–2003) and P75 threshold are identical to the historical analysis.

Interpretation. The current MFLS signal (49.5) is above the P75 alarm threshold (24.0) but at a modest $+0.60\sigma$ Z-score, indicating **elevated but not critical** systemic risk. The 4-quarter trend is falling (-1.8%), suggesting the system is moving toward lower stress. The CCyB recommendation of 180 bps (down from 192 bps in 2024-Q1) reflects this gradual easing. The herding score of 0.42 (moderate) indicates that institutional behaviour diversity is neither collapsed (convergent herding) nor extreme (idiosyncratic risk).

The 71-quarter GFC "lead" requires careful interpretation: it does *not* mean the model predicted the GFC 17 years in advance. Rather, it reveals that the U.S. banking system has operated in a *persistently super-critical* state—above the P75 normal-period threshold—for most of the 1990–2025 sample. This is consistent with the paper's finding that 68.6% of quarters lie above the critical manifold $\mathcal{C}$ (Section 8). The alarm identifies a *regime* (super-criticality) rather than a specific event; the GFC was a phase transition *within* this regime.

Table 10: Live prediction results: U.S. banks (N=20, T=144, d=5). All data fetched March 2026; latest quarter is 2025-Q4. P75 threshold trained on pre-2007 data.

Metric	U.S. Banks (N=20)
AUROC	0.577
95% CI	[0.364, 0.807]
Hit Rate	82.6%
False Alarm Rate	62.8%
GFC Lead	71Q†
Spectral Radius	7.67
Peak CCyB	250 bps
Granger (best p)	0.452
Current state (2025-Q4):	
MFLS Signal	49.5
Alarm Threshold	24.0
Above Threshold	Yes
Signal Z-score	$+0.60\sigma$
CCyB	180 bps
4Q Trend	Falling (-1.8%)
Herding Score	0.42 (moderate)

† 71Q reflects persistent super-criticality: the system has been above the P75 threshold almost continuously since the early 1990s.

G-SIB panel. The global panel (N=24 institutions from 11 countries across 5 regions—US, UK, EU, Asia, Africa—T=88 quarters, 2003–2024) achieves AUROC = 0.811 with 95% CI $[0.662, 0.982]$, hit rate 95.7%, and 19-quarter GFC lead time. Data sources: FDIC SDI (US banks), World Bank API v2 (all regions), and ECB MIR (Eurozone funding costs). Nigerian banks (Zenith, GTBank, First Bank, Access, UBA) use World Bank country-level indicators only.

Regional interpretation. The US panel (N=5, FDIC bank-level data) produces the most granular signals and lowest false-alarm rate; its Z-score of -0.52σ indicates the US banking system currently sits *below* its historical mean stress—consistent with the detailed analysis (Section 8). The UK (HSBC, Barclays), EU (BNP Paribas, Société Générale, Deutsche Bank, ING, UniCredit), and Asia (MUFG, Mizuho, SMFG, ICBC, CCB, Bank of China, Agricultural Bank of China) use country-level World Bank indicators, which amplify cross-country dispersion. Nigeria's five major banks achieve the highest regional AUROC (0.893), reflecting the sharp structural breaks in Nigerian banking data around the 2009 consolidation reforms and 2016 recession.

The institution-level audit identifies ING (Z=$+4.60\sigma$, dominant channel δ_C) as the principal outlier, driven by idiosyncratic Dutch credit-cycle positioning.

Granger causality (live data). All three Granger tests on live U.S. data fail at $\alpha = 0.10$ (best $p = 0.452$), confirming the structural finding: the MFLS signal tracks a phase-transition

Table 11: Global multi-region MFLS analysis (N=24, T=88, d=5). Per-region engines fitted with identical calibration (2003–2006 normal period).

Region	N	Signal	Threshold	Z	AUROC
US	5	30.7	16.4	-0.52σ	0.400
UK	2	$3.5 \times 10^{6\dagger}$	969	$+5.26\sigma$	0.835
EU	5	1 557	20.2	$+4.61\sigma$	0.666
Asia	7	119 432	134	$+5.26\sigma$	0.658
Africa	5	4 912	50.1	$+5.44\sigma$	0.893
Global	**24**	**639**	**35.0**	$\mathbf{+4.05\sigma}$	**0.811**

† Country-level World Bank aggregates inflate cross-country Mahalanobis distance. Regime classification (CRITICAL vs. NORMAL) is more meaningful than raw signal comparison across regions.

process, not a linear trend. Linear Granger (global panel, best $p = 0.333$) and threshold Granger ($p = 0.523$) also fail, indicating that this structural property generalises across regions.

Multi-variant robustness on the global panel. Table 12 reports all five MFLS scoring variants on the same 24-institution global panel. Supervised variants (QuadSurf, SignedLR, ExpoGate) are trained on the first 70% of the timeline (2003–2018, including the GFC), with detection evaluated over the full period.

Table 12: Five MFLS variants on the global panel (N=24, T=88). Supervised variants (†) trained on 70% of timeline. HR/FAR at P75 threshold.

Variant	AUROC	HR	FAR	Lead	Current Z
Baseline (Mahalanobis)	**0.808**	91.3%	38.5%	19Q	$+4.11\sigma$
Full-BSDT (4-ch)	0.785	60.9%	20.0%	17Q	$+5.46\sigma$
SignedLR†	0.795	60.9%	30.8%	15Q	$+1.37\sigma$
QuadSurf†	0.531	39.1%	16.9%	7Q	-0.71σ
ExpoGate†	0.531	39.1%	16.9%	7Q	-0.73σ

Variant insights. Three findings emerge from the multi-variant analysis:

1. **No single variant dominates all regions.** FullBSDT (unsupervised, 4-channel uniform sum) is best for Asia and EU, where country-level data makes feature-gap and temporal-novelty channels informative. ExpoGate wins in Africa (AUROC 0.913) where Nigeria's structural breaks suit the sigmoid gating. SignedLR is best for UK (0.857), and QuadSurf/ExpoGate dominate the US (0.841), where polynomial features capture nonlinear interactions invisible to the Mahalanobis baseline.

2. **SignedLR channel weights reveal regional risk drivers.** The global SignedLR learns: $\beta_{\delta_T} = +1.91$ (temporal novelty), $\beta_{\delta_G} = +0.84$ (feature gap), $\beta_{\delta_C} = +0.52$ (credit camouflage), $\beta_{\delta_A} = -1.47$ (activity anomaly). The negative δ_A weight is notable: crises

Table 13: Per-region × per-variant AUROC matrix. Each variant fitted independently per region with identical normal-period calibration (2003–2006). Best per region in **bold**.

Region	Baseline	FullBSDT	QuadSurf†	SignedLR†	ExpoGate†
Africa	0.768	0.833	0.909	0.898	**0.913**
Asia	0.659	**0.821**	0.701	0.375	0.701
EU	0.666	**0.722**	0.457	0.341	0.457
UK	0.834	0.822	0.666	**0.857**	0.673
US	0.451	0.364	**0.841**	0.826	**0.841**
Global	**0.808**	0.785	0.531	0.795	0.531

are preceded by *reduced* velocity-anomaly, not heightened activity—consistent with the "calm before the storm" phenomenon. At the regional level, Africa's crisis signal is dominated by δ_C (+4.17), reflecting credit-driven instability in Nigerian banking, while the UK is driven by δ_G (+2.22), reflecting structural feature gaps.

3. **QuadSurf and ExpoGate correlate at $r = 0.998$.** The tanh/sigmoid gating of ExpoGate adds negligible information beyond the polynomial surface, confirming that the degree-2 polynomial captures the necessary nonlinearity. Both supervised surface variants are *anti-correlated* with the unsupervised Baseline ($r \approx -0.34$), meaning they detect different aspects of systemic stress—the Baseline tracks distance from equilibrium while supervised variants track probability of transition.

9 Welfare Calibration

9.1 CCyB Calibration Formula

Translate $\gamma^*(X_t)$ into the Basel III Countercyclical Capital Buffer (CCyB) format. Under a log-linear approximation around steady state, the required buffer increment is:

$$\Delta\text{CCyB}(t) \approx \kappa^{-1} \cdot \gamma^*(X_t) \cdot \sigma_\ell(t), \tag{9}$$

where $\sigma_\ell(t)$ is the cross-sectional standard deviation of leverage in the banking system at time t, observable quarterly from BIS data. Equation (9) converts the abstract coupling coefficient into basis points, directly comparable to the Basel III CCyB range of 0–250 bps.

9.2 Welfare Cost of Inaction

The expected welfare loss from running at $\gamma = 0$ over an instability window $[t_0, t_1]$ is:

$$\mathcal{L}_{\text{inaction}} = \int_{t_0}^{t_1} \left[E(X_t) - E(X_t^{\gamma^*})\right] dt.$$

Converting to consumption-equivalent units via a standard Euler equation ($C_t \propto e^{-E(X_t)/(\sigma\theta)}$), the percentage loss in permanent consumption is:

$$\%CL = 100 \times \left(1 - e^{-\mathcal{L}_{\text{inaction}}/(\sigma\theta T)}\right).$$

Table 14 reports the calibrated welfare losses for the three crisis windows on real FDIC data. The GFC 2008 window ($T = 10$ quarters of elevated energy above the median) produces a consumption-equivalent loss of 19.57 percentage points, consistent with the consensus range of macro studies [14]. COVID 2020 shows near-zero welfare loss in the model because the energy spike dissipated within two quarters under emergency policy response; the CCyB path nevertheless peaks at 231 bps, reflecting the severity of the initial shock. The 2022 rate-shock episode produces a modest loss of 1.22 pp, consistent with a stress event absorbed without systemic transmission.

Table 14: Welfare calibration results on real FDIC data ($T = 140$, $N = 7$, $d = 6$). $\mathcal{L}$: consumption-equivalent welfare loss (percentage points). Peak CCyB: maximum implied buffer increment (basis points). See Section 9 for calibration details.

Crisis Window	$\mathcal{L}$ (pp)	Peak CCyB (bps)
GFC 2008 (2007-Q3–2009-Q4)	19.57	152
COVID 2020 (2019-Q4–2021-Q2)	0.00	231
Rate Shock 2022 (2022-Q1–2023-Q4)	1.22	0

Notes. COVID $\mathcal{L} = 0$ because no quarter in that window exceeded the reference-period median plus two consecutive quarters of above-threshold energy; the unprecedented policy response compressed the crisis duration below the model's persistence threshold. The COVID CCyB peak of 231 bps reflects that the *fragility level* was extreme even though the welfare loss was averted. The 2022 CCyB = 0 bps reflects that $\gamma^*(t)$ was near zero in that window: the rate shock was absorbed by institution-level adaptation without a systemic energy spike.

9.3 Pigouvian Interpretation

From Section 2, the optimal tax is $\tau_i^* = \gamma^*(X)\,\partial E/\partial \ell_i$. The macro-prudential instrument is therefore a *gradient-indexed levy*: the surcharge on institution i is proportional to how much its marginal leverage change increases the system energy. Institutions with high connectivity (large $\partial E/\partial \ell_i$) pay more; institutions in isolation pay less. The total tax $\gamma^*(X)\,\|\nabla E(X)\|^2$ equals the welfare gap $V_{\bar{\gamma}} - V^*$ to leading order, confirming the Pigouvian interpretation: the tax exactly corrects the externality.

10 Discussion

10.1 Summary of Empirical Contributions

The empirical analysis on FDIC SDI call-report data ($T = 140$, $N = 7$, $d = 6$) produces six findings:

1. **Super-criticality is the norm.** The U.S. banking system spends 68.6% of the 1990–2024 sample above the critical manifold $\mathcal{C}$. Crises do not push the system above $\mathcal{C}$; they exploit a system already in a super-critical state.

2. **Anti-alignment reveals complementarity.** The gradient alignment between E_{BS} and Φ is $\cos\theta = -0.335$ on real data. This negative alignment indicates that detection (∇E_{BS})

and restoration ($-\nabla\Phi$) measure *geometrically perpendicular* axes of instability, making their combination more informative than either alone.

3. **6-quarter GFC lead.** The MFLS alarm fires in 2007-Q1 using only a threshold trained on pre-2007 data—six quarters before Lehman, with a 71.4% hit rate and no false alarms in the pre-crisis training period.

4. **Universal Granger failure.** Five independent MFLS variants—spanning unsupervised, polynomial, logistic, and gated families—all fail Granger causality at $\alpha = 0.10$. The failure is structural: crises are phase transitions, not trends, and linear temporal predictability is the wrong test.

5. **Precision-recall frontier.** Variants trade off monotonically between hit rate (71.4%–14.3%) and selectivity (1.9×–4.3×), tracing a Pareto curve that separates "distance from equilibrium" from "probability of transition."

6. **Herding discovery.** The Signed LR variant learns a negative weight ($\beta_{\delta_T} = -1.38$) on temporal novelty: before crises, institutions become *more* predictable from their own history. This is the quantitative signature of convergent herding—the channel most associated with systemic risk is not deviation from the norm, but *conformity to it*.

7. **Live prediction (2025-Q4).** On the latest FDIC data (through 2025-Q4, N=20 individual banks), the MFLS signal stands at 49.5 ($+0.60\sigma$), above the P75 alarm threshold, with a falling trend (-1.8% over four quarters) and moderate herding (0.42). The CCyB recommendation has declined from 192 bps (2024-Q1) to 180 bps, consistent with a system that remains persistently super-critical but is not approaching a phase-transition boundary. All Granger tests continue to fail (best $p = 0.452$) on the extended sample.

8. **Global multi-region analysis.** A 24-institution panel spanning US (5), UK (2), EU (5), Asia (7), and Africa/Nigeria (5) achieves AUROC = 0.811 [0.662, 0.982] with 19-quarter GFC lead and 95.7% hit rate. Nigeria's banking sector contributes the highest regional AUROC (0.893), reflecting sharp structural breaks around the 2009 consolidation reforms. The δ_C (credit camouflage) channel dominates across all non-US regions; ING is the sole institution-level outlier at $+4.60\sigma$.

10.2 Limitations

First-order dynamics. GravityEngine implements first-order gradient flow; it has no velocity state or mass matrix. The proofs in this paper are calibrated to this model. An extension to second-order (inertial) dynamics would require replacing the Lyapunov candidate $V = \Phi(X)$ with $V = \Phi(X) + \frac{1}{2}\dot{X}^\top M\dot{X}$ and re-deriving the stability rate.

Homogeneous feature space. All institutions are modelled in the same d-dimensional space. Heterogeneous institution types (commercial banks vs. insurance companies vs. hedge funds) have different natural feature spaces. The framework extends by using a block-diagonal covariance Σ_0 and separate deviation operators per class, but the equivalence constants c_1, c_2 will depend on the block structure.

Static network. The critical manifold $\mathcal{C}$ depends on $\lambda_{\max}(\mathbf{W})$, the spectral radius of the network. The results assume a fixed $\mathbf{W}$; in practice, the network topology changes over time (especially during crises). Extending Theorem 5.3 to a time-varying $\mathbf{W}(t)$ is an open problem.

Panel coverage. The institution-level panels ($N = 20$ U.S. banks, $N = 24$ global institutions across 5 regions) exclude banks with $< 25\%$ data coverage. The global panel uses country-level World Bank indicators rather than bank-level data for non-US institutions, meaning all banks from the same country share identical features. This inflates cross-country Mahalanobis distance and makes per-region signal magnitudes non-comparable without normalisation. Nigerian banks (added as an emerging-market benchmark) use World Bank data only, with no ECB or FDIC granularity.

Granger causality. The universal Granger failure, while theoretically consistent with the phase-transition thesis, means that the model cannot be validated by standard linear forecasting metrics. Developing nonlinear or threshold-based predictability tests that align with the theoretical framework is an important next step.

10.3 Open Problems

1. **Global Lyapunov extension.** Theorems 5.2 and 5.3 are local (linearisation-based). A global Lyapunov argument for the full nonlinear system above $\mathcal{C}$ would require controlling higher-order terms in the Gronwall bound—the force-clamp (Appendix A8) provides the regularisation needed, but the argument is not yet complete.

2. **Heterogeneous equilibria.** The uniqueness of $X^\star$ is not proved; the erf-log pairwise potential is non-convex and admits multiple local minima. Understanding the basin structure is important for policy: which equilibrium the system converges to after a crisis depends on initial conditions.

3. **Tighter separation bounds.** Proposition 4.2 derives $\nu > 0$ from the condition number $\kappa(\Sigma_0)$ of the reference covariance. The current bound $\nu_C \geq \kappa(\Sigma_0)^{-1}$ is conservative; tighter bounds incorporating the distribution of eigenvalues (not just extremes) and the interaction between operators could yield sharper equivalence constants c_1, c_2 in Theorem C.

4. **Nonlinear forecasting tests.** Replace Granger causality with tests designed for threshold/phase-transition processes: exceedance regression [10], quantile causality, or time-varying parameter models that can detect state-dependent predictive power.

5. **Real-time estimation.** The current analysis uses full-sample Ledoit–Wolf shrinkage. Deploying the framework in real time requires recursive covariance estimation and rolling BSDT re-calibration with a specified look-back window; the computational cost is $O(N^2 d)$ per quarter, which is operationally feasible.

10.4 Policy Implications

For central banks to adopt the CCyB formula (9) in practice, three conditions must be met: (1) the network adjacency matrix $\mathbf{W}$ must be observable with quarterly frequency (currently

available from BIS bilateral exposures for 30 countries); (2) the deviation operators δ_k must be re-calibrated annually on updated normal-period data (computationally tractable at $O(N^2)$ per re-calibration); and (3) the intervention cost κ must be estimated from GDP data (the ratio of growth lost per unit of credit restriction). All three are operationally feasible with existing BIS and FRED data infrastructure.

The herding discovery (finding 6) has a direct policy implication: a macro-prudential indicator should weight *decreasing* behavioural diversity as a warning sign, not just increasing distance from equilibrium. The current Basel III credit-to-GDP gap measures only aggregate leverage deviation; the MFLS framework adds cross-sectional *convergence* as a crisis predictor.

Appendices

A Full Proofs of Theorems 1–6 and C

We provide complete proofs for all theorems in the main text. Proofs are ordered by increasing generality.

Proof of Theorem 5.1 (Monotone Descent, full). Define $V(t) = \Phi(X(t))$. Differentiating: $\dot{V} = \nabla\Phi(X)^\top \dot{X} = \nabla\Phi^\top(-\nabla\Phi) = -\|\nabla\Phi\|^2 \leq 0$. Since $V(t)$ is non-increasing and bounded below by $\Phi(X^\star)$, it converges to $V_\infty \geq \Phi(X^\star)$. By LaSalle, every ω-limit point X_∞ satisfies $\nabla\Phi(X_\infty) = 0$. *Discrete case*: backtracking line search in `gravity.py` (lines 659–670) explicitly rejects steps that increase energy; the force clamp at $F_{\max} = 100$ prevents numerical overflow. □

Proof of Theorem 5.2 (Local Exponential Stability, full). Write $z(t) = X(t) - X^\star$ with $\|z\|$ small. Expanding: $\dot{z} = -\nabla\Phi(X^\star + z) = -H^\star z + O(\|z\|^2)$. Since $H^\star \succ 0$ (Assumption E2), all eigenvalues of $A^\star = -H^\star$ have real part $-\lambda_i(H^\star) < 0$. By Lyapunov's indirect method, $X^\star$ is locally exponentially stable with rate $r = \lambda_{\min}(H^\star)$.

Gronwall bound: decompose $\dot{z} = -\alpha z + \Delta_{\text{pair}}$ with $\|\Delta_{\text{pair}}\| \leq L_{\text{pair}}\|z\|$. Then: $\frac{d}{dt}\|z\|^2 \leq -2\alpha\|z\|^2 + 2L_{\text{pair}}\|z\|^2 = -2(\alpha - L_{\text{pair}})\|z\|^2$. For $\kappa = \alpha - L_{\text{pair}} > 0$, $\|z\|^2(t) \leq \|z\|^2(0)e^{-2\kappa t}$. □

Proof of Theorem 5.3(ii) (Growing Mode, full). Above $\mathcal{C}$, let u be the unit eigenvector of $D^2\Phi_{\text{pair}}(X^+)$ corresponding to $\lambda_u > \alpha$. Define $W(t) = \frac{1}{2}\langle X(t) - X^\star, u\rangle^2$. Differentiating: $\dot{W} = \left\langle \dot{X} - 0, u \right\rangle \langle X - X^\star, u\rangle = \left\langle -\alpha(X - X^\star) - D^2\Phi_{\text{pair}}(X - X^\star) + O(\|z\|^2), u \right\rangle \langle X - X^\star, u\rangle$. Projecting onto u: $\dot{W} = (-\alpha + \lambda_u)W + O(\|z\|^3) = cW + O(\|z\|^3)$ with $c = \lambda_u - \alpha > 0$. By forward Gronwall: $W(t) \geq W(0)e^{ct/2}$ for small t. This grows exponentially; no fixed $\bar{\gamma}$ prevents it (increasing $\bar{\gamma}$ shifts $\mathcal{C}$ but there always exist configurations above the shifted $\mathcal{C}$ for small enough pairwise distances). □

Proof of Proposition 6.1 (OGD Regret, full). Let $\Delta_\gamma = \gamma_{\max} - \gamma_{\min}$. Standard OGD [3] with convex c_t and step $\eta_\gamma = \Delta_\gamma/\sqrt{T}$: $R_T \leq \frac{\Delta_\gamma^2}{2\eta_\gamma} + \frac{\eta_\gamma}{2}\sum_t \|\nabla_\gamma c_t\|^2 \leq \frac{\Delta_\gamma\sqrt{T}}{2} + \frac{L_c^2\Delta_\gamma T}{2\sqrt{T}} = L_c\Delta_\gamma\sqrt{T}$.

Oracle approximation by power iteration: maintain unit vector v_t; update $\tilde{v}_{t+1} = (\nabla\Phi_{\text{pair}}(X^t + hv_t) - \nabla\Phi_{\text{pair}}(X^t))/h$, renormalize. After K steps, the Rayleigh quotient $\hat{\lambda}_t$ satisfies $|\hat{\lambda}_t - \lambda_{\max}| \leq \lambda_{\max}(\lambda_2/\lambda_{\max})^{2K}$. Choosing $K = O(\log N)$ gives approximation error $O(N^{-1})$. □

Proof of Theorem 7.2 (Optimal Policy, full derivation). First-order condition of the Bellman equation w.r.t. γ: $0 = 2\kappa\gamma + \beta\,\partial_\gamma \mathbb{E}[V(X') \mid X, \gamma]$. Using $\partial_\gamma X' = -\eta \nabla E(X)$ and $\partial_{X'} V \approx \nabla E(X')/(1-\beta)$ (first-order approximation): $\partial_\gamma \mathbb{E}[V] \approx -\eta/(1-\beta)\,\langle \nabla E(X'), \nabla E(X)\rangle$. At leading order $\nabla E(X') \approx \nabla E(X)$, giving $2\kappa\gamma^* \approx \beta\eta\,\|\nabla E\|^2/(1-\beta)$, so $\gamma^* \approx \frac{\beta\eta\|\nabla E\|^2/(1-\beta)}{2\kappa}$. The saturation factor $E/(E+\theta)$ with $\theta = \kappa/\beta$ ensures $\gamma^* \to 0$ as $E \to 0$ (no intervention needed in the normal regime). □

Proof of Lemma (Force-Clamp Regularisation – Appendix to Proof 7). The clamped force $F_{\text{clamp}} = \Pi_{B(F_{\max})}(F_{\text{raw}})$ is the projection onto the closed ball of radius $F_{\max}$. Projection onto a convex set is non-expansive: $\|F_{\text{clamp}}(X) - F_{\text{clamp}}(Y)\| \le \|F_{\text{raw}}(X) - F_{\text{raw}}(Y)\|$. On any sublevel set $\mathcal{L}_c$, log-repulsion coercivity gives minimum pairwise distance $\delta_{\min}(c) > 0$, so the Jacobian norm of F_{raw} is bounded by $F_{\max}/\delta_{\min}(c) < \infty$. The contraction condition $\alpha > F_{\max}/\delta_{\min}(c)$ is achievable for any finite c by increasing α. □

B Computational Details

B.1 GravityEngine Parameter Table

Parameter	Symbol	Default value	Role
Radial spring	α	0.1	Mean-reversion strength
Pairwise coupling	γ	1.0	Interaction scale
Gaussian range	σ	1.0	Attraction length-scale
Repulsion coefficient	λ	0.1	Prevents collapse
Softening	ε	10^{-5}	Regularises log-repulsion
Force clamp	$F_{\max}$	100	Bounds pairwise Jacobian
Step size	η	0.02	Euler step
Backtracking factor	—	0.5	Armijo line-search halving

B.2 Pairwise Force Kernel

The erf-based pairwise potential between agents i and j at distance $r = \|x_i - x_j\|$ is:

$$\phi(r) = \gamma \frac{\sigma\sqrt{\pi}}{2} \operatorname{erf}\left(\frac{r}{\sigma}\right) - \gamma\lambda \log(r + \varepsilon).$$

The attraction force $-\phi'(r)\hat{r}$ is Gaussian: $\phi_1'(r) = \gamma e^{-r^2/\sigma^2}$. The repulsion force is long-range logarithmic: $\phi_2'(r) = \gamma\lambda/(r+\varepsilon)$. The net force is attractive at long range and repulsive at short range, giving a stable cluster configuration at the equilibrium distance $r^* = \sigma\sqrt{\log(1/\lambda)}$ (implicit solution of $\phi'(r^*) = 0$).

B.3 Finite-Difference HVP

The Hessian-vector product $D^2\Phi_{\text{pair}}(X)v$ required by the power iteration is computed via the symmetric finite difference:

$$D^2\Phi_{\text{pair}}(X)v \approx \frac{\nabla\Phi_{\text{pair}}(X + hv) - \nabla\Phi_{\text{pair}}(X - hv)}{2h}, \quad h = 10^{-4}.$$

This requires two gradient evaluations ($O(N^2 d)$ each) per power iteration; the approximation error is $O(h^2 \left\| D^3\Phi \right\|)$, negligible for $h = 10^{-4}$.

C Data Sources and Processing

C.1 FDIC Statistics on Depository Institutions (SDI)

All banking features (f_0–f_4) are drawn from the FDIC SDI quarterly call reports, accessed via the public REST API at `https://banks.data.fdic.gov/api/financials`. Data are aggregated by `SPECGRP` (institution type: 7 categories) and queried at individual quarter-end dates (e.g., `filters=REPDTE:20070331`). The six call-report fields used are: `ASSET` (total assets), `LNLSNET` (net loans and leases), `EQ` (total equity capital), `NCLNLS` (noncurrent loans and leases), `NETINC` (net income), `EINTEXP` (total interest expense—note: this is the correct FDIC field name, not `INTEXP`).

C.2 Federal Reserve FRED Series

Feature f_5 (yield-curve slope) is the 10-Year minus 2-Year Treasury Constant Maturity spread (series T10Y2Y), downloaded from FRED at quarterly frequency.

Benchmark stress indices used for lead-lag analysis:

- STLFSI: St. Louis Fed Financial Stress Index (weekly, averaged to quarterly)
- VIX: CBOE Volatility Index (daily close, averaged to quarterly)
- NFCI: Chicago Fed National Financial Conditions Index (weekly, averaged to quarterly)
- T10YFF: 10-Year Treasury minus Fed Funds Rate (spread used in SRISK-proxy composite)

C.3 Normal-Period Definition

The normal reference period is 1994-Q1 through 2003-Q4 (40 quarters). This span is chosen to be: (a) post-S&L resolution (completed 1993); (b) pre-housing boom (credit growth accelerated 2004); (c) long enough for stable covariance estimation ($T_{\text{ref}} = 40 \gg d = 6$). All BSDT operator parameters (μ_0, Σ_0, PCA loadings, velocity thresholds, KDE bandwidths) are fitted on this reference period only.

C.4 Replication

The full replication package, including all data-loading scripts, pipeline code, figure-generation scripts, and cached API responses, is available at `https://github.com/segetii/fintech` (branch `sync/merge-2025-10-22`). The pipeline can be reproduced with:

```
cd adaptive-friction-stability-upgraded/pipeline
python run_pipeline.py
```

D Additional Crisis Window: 2022 Rate Shock

The 2022 rate-shock window (2022-Q1 through 2023-Q4) is characterised by the Federal Reserve's 525-basis-point policy rate increase, the fastest tightening cycle since 1980. The MFLS signal shows a moderate energy spike beginning 2022-Q2, but crucially, the OOS alarm (trained on P75 of pre-2007 data) does *not* fire during this window: the energy level remains below the GFC-calibrated threshold.

This is consistent with the model: the 2022 stress was publicly anticipated via forward guidance, allowing institutions to adapt. The welfare loss is modest ($\mathcal{L} = 1.22$ pp) and the implied CCyB is zero. The model correctly distinguishes between a predictable repricing event (2022) and an abrupt structural crisis (2008).

References

[1] Adrian, T. and Brunnermeier, M.K. (2016). CoVaR. *American Economic Review*, 106(7), 1705–1741.

[2] Brownlees, C. and Engle, R.F. (2017). SRISK: A conditional capital shortfall measure of systemic risk. *Review of Financial Studies*, 30(1), 48–79.

[3] Hazan, E. (2016). *Introduction to Online Convex Optimization*. Foundations and Trends in Optimization, 2(3–4), 157–325.

[4] Khalil, H.K. (2002). *Nonlinear Systems*, 3rd ed. Prentice Hall.

[5] LaSalle, J.P. (1960). Some extensions of Liapunov's second method. *IRE Transactions on Circuit Theory*, 7(4), 520–527.

[6] Ledoit, O. and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2), 365–411.

[7] Lohmiller, W. and Slotine, J.-J.E. (1998). On contraction analysis for non-linear systems. *Automatica*, 34(6), 683–696.

[8] Trefethen, L.N. and Bau, D. (1997). *Numerical Linear Algebra*. SIAM.

[9] Acemoglu, D., Ozdaglar, A. and Tahbaz-Salehi, A. (2015). Systemic risk and stability in financial networks. *American Economic Review*, 105(2), 564–608.

[10] Allen, L., Bali, T. and Tang, Y. (2012). Does systemic risk in the financial sector predict future economic downturns? *Review of Financial Studies*, 25(10), 3000–3036.

[11] Bertsekas, D.P. (2012). *Dynamic Programming and Optimal Control*, 4th ed. Athena Scientific.

[12] Brunnermeier, M.K. (2009). Deciphering the liquidity and credit crunch 2007–2008. *Journal of Economic Perspectives*, 23(1), 77–100.

[13] Elliott, M., Golub, B. and Jackson, M.O. (2014). Financial networks and contagion. *American Economic Review*, 104(10), 3115–3153.

[14] Greenwald, D.L., Jordà, O., Schularick, M. and Taylor, A.M. (2020). Financial crisis recoveries. *AEA Papers and Proceedings*, 110, 124–129.

[15] May, R.M., Levin, S.A. and Sugihara, G. (2008). Complex systems: Ecology for bankers. *Nature*, 451, 893–895.

[16] Shalev-Shwartz, S. (2012). Online learning and online convex optimization. *Foundations and Trends in Machine Learning*, 4(2), 107–194.